Training-Free Ultra Small Model for Universal Sparse Reconstruction in Compressed Sensing

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***Abstract*—Despite large models drive unprecedented growth in data and model parameters, many real-world problems prioritize interpretability and generality, and lack sufficient training data. For instance, in Compressed Sensing (CS) where sparse reconstruction solves underdetermined systems, traditional iterative methods remain the practical choice due to their interpretability and out-of-the-box applicability to arbitrary conditions, but suffer from poor quality and inefficiency at low sampling rates. To address this, we propose Coefficients Learning (CL), a novel training-free framework for sparse reconstruction. CL employs ultra-small neural models with only n trainable parameters for a length-n signal. It retains the interpretability and generality of traditional iterative methods by adopting their residual-based solving process, while enhancing efficiency and accuracy by replacing closed-form solutions with automatic differentiation and embedding prior knowledge into the model losses. We evaluate CL extensively on synthetic and real one-dimensional and two-dimensional signals. A detailed analysis is first conducted using an implemented CLOMP. To demonstrate general applicability, CL is also implemented on three types of classic iterative CS reconstruction methods. Results show that CL maintains the generality of iterative methods while significantly boosting accuracy. Although it adds minor overhead for convex optimization or message-passing methods, it achieves efficiency gains of 100 to 1000 times for greedy algorithms. On the tested nine diverse image datasets, CL improves median reconstruction accuracy by approximately 163%, 78%, and 35% at sampling rates of 0.04, 0.25, and 0.5, respectively, compared to classic iterative methods. This training-free CS reconstruction method can truly empower countless industrial or medical machines that rely on sparse solution.**

***Index Terms*—Training-free neural network, compressed sensing, sparse reconstruction, coefficients learning.**

I. Introduction

RECENT advances in largescale models have pushed the volume of data and parameters to new heights. However, in many realworld physical applications, acquiring sufficient training data remains difficult, and interpretability and generality are highly valued [1], [2]. For example, in solving underdetermined problems, feeding large amounts of data still cannot guarantee the general applicability of AI methods, making their adoption unreliable in areas such as computational medical imaging and lowpower sparse sensing. Moreover, larger models impose greater resource requirements on deployment platforms. This paper studies AIbased general sparse reconstruction for Compressive Sensing (CS), a typical underdetermined problem. We explore an approach to enhance the practicality of AI models by drawing on the interpretability of traditional methods rather than merely accumulating data and parameters.

CS is a well-proved theory that integrates sparse representation and sparse reconstruction. Since formally proposed in 2006 [3], [4], CS has broken the nearly century-long reign of the Nyquist sampling theorem with rigid mathematical proof, leading to various breakthroughs in fields such as engineering imaging with the idea of single pixel imaging [5], [6], [7], [8], [9], and biomedical imaging like stochastic optical reconstruction microscopy [10], fluorescence imaging [11] and magnetic resonance imaging [12], genetic engineering [13], optical imaging [14], [15], quantum research [16], etc. CS significantly reduces sensing time, data volume, and brings other advantage like robust to partial data loss. However, CS suffers from a time- and computationally expensive sparse reconstruction process, which limits its application in many real-world scenarios. Some research efforts have focused on developing specialized hardware to accelerate reconstruction, such as in-memory analog solution [17]. The current three categories [18] (Iterative reconstruction, end-to-end learning, deep unfolding) of CS/sparse reconstruction methods fail to fully consider the generality, interpretability, efficiency, and accuracy. This is also a challenge for countless under-determined systems that rely on sparse solution.

Received 23 March 2025; revised 12 January 2026; accepted 28 March 2026. Date of publication 2 April 2026; date of current version 6 July 2026. This work was supported in part by National Natural Science Foundation of China under Grant 62573206 and Grant 62103154, in part by Young Elite Scientists Sponsorship Program by CAST under Grant 2023QNRC001, in part by National Science Fund for Distinguished Young Scholars of China under Grant 62225603, and in part by Key International Cooperation Research Projects of NSFC under Grant 61960206010. Recommended for acceptance by J. Han. *(Corresponding authors: Chaoqing Tang; Xiang Bai.)*

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The data and code for this paper are available at https://github.com/BillTtzqgbt/CSCoefficientsLearning.git

Digital Object Identifier 10.1109/TPAMI.2026.3680162

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We observe that many iterative sparse reconstruction algorithms can be decomposed into two core blocks. The first extracts new atom information from the residual and injecting it into the new estimation, thereby satisfying the L0-norm constraint. The second then updates the sparse coefficients (denoted as $\mathbf{s}$) and the residual via a computationally intensive equation.

Based on the above finding, this paper proposes ultra-small artificial neural models called Coefficients Learning (CL), enabling training-free and rapid sparse reconstruction while perfectly inheriting the generality and interpretability of traditional iterative methods. The small model here is relative to the parameter number of current AI methods for solving CS sparse reconstruction. The perfect generality, interpretability, and good efficiency, accuracy enable CL to effectively replace traditional iterative algorithms for sparse reconstruction. This is particularly beneficial for applications where reliability is important and training data is limited, such as resource-limited sensors in wireless sensor networks, implantable biomedical devices, and medical imaging. Since compressed sensing is a typical under-determined linear system, the proposed CL method also works for countless applications that require sparse solution of under-determined linear systems, such as image super-resolution, modeling of electromagnetic scattering, and blind signal separation.

The major contributions of this work are:

1) We propose a new category of CS reconstruction methods called Coefficients Learning, enabling training-free and rapid sparse reconstruction while perfectly inheriting the generality and interpretability of traditional iterative methods, bringing additional advantage of incorporating prior knowledge to improve the reconstruction quality. Current AI-based CS reconstruction methods often fail when there are changes in sampling rates, measurement matrices, signal dimensions, or applications that have limited training data.
2) We propose the overall structure of CL. In CL, a signal of length n only needs a minimal of n trainable parameters. For example, for 33×33 image patches in CS image reconstruction, the proposed CL method uses only 0.5% and 0.6% of the parameters compared to the state-of-the-art Deep Unfolding methods AMPNet and ISTANet, respectively, which is ultra small and fast.
3) We first implement CL on the popular orthogonal matching pursuit (OMP) as CLOMP for performance analysis. To validate the applicability to other iterative reconstruction methods, more CL methods are implemented on three types of classic iterative reconstruction methods. Experiments are conducted on various one-dimensional (1D) signals and two-dimensional (2D) signals.
4) The efficiency of CL implementation on greedy algorithms like CLOMP are 2$\sim$3 orders of magnitude higher than that of traditional iterative algorithms for large-scale data.
5) On the tested nine diverse image datasets, the implemented four CL methods improves median reconstruction accuracy by approximately 163%, 78%, and 35% for $Sr = 0.04$, 0.25, and 0.5, respectively while maintaining the generality.

II. Related Works

A. Iterative Reconstruction Methods

The first category is the traditional iterative reconstruction methods. Three representative sub-categories [19], [20] are greedy algorithms, such as Matching Pursuit (MP), Orthogonal Matching Pursuit (OMP) and Iterative Hard Thresholding (IHT), convex optimization techniques, such as the Iterative Soft Thresholding Algorithm (ISTA) and the Alternating Direction Method of Multipliers (ADMM), message passing, such as Approximate Message Passing (AMP). These methods are well-supported by mathematical proof [21], [22], i.e., perfect interpretability. They also exhibit strong generality, as they only require sparsity as a priori knowledge, rather than relying on a training dataset or a fixed measurement matrix. However, a significant drawback is their low efficiency, particularly when dealing with large-scale data, such as long sequences or high-resolution images. Since CS theory is primarily designed for one-dimensional signal, for example, when transfer a 256×256 image into 1D sequence using OMP can take several hours on a consumer-grade CPU.

B. End-to-End Learning

The second category is end-to-end learning [23], [24], [25], [26], [27], which involves constructing a neural network model that directly maps the CS measurement $\mathbf{y}$ to the target signal $\mathbf{x}$. With the thriving of deep learning [28], the neural network model can take various forms, including fully connected networks [29], convolutional neural networks [11], [25], [30], [31], recurrent neural networks [32], [33], generative adversarial networks [34], [35], Transformer [36], etc. These methods rely on a large training dataset for effective end-to-end training, so the generality is poor on a different dataset or application. The reconstruction efficiency and accuracy are good after training, but the overall efficiency remains a concern, as the training process is time-consuming and can be less efficient than traditional iterative reconstruction methods. Additionally, these deep learning models are totally opaque systems, which has poor interpretability and reliability [37].

C. Deep Unfolding

The third category is Deep Unfolding that thrives from 2016 [38], [39] and remains a hot topic in recent research [32], [40], [41], [42], which enhances the interpretability. These methods construct neural network layers to simulate the iterative rounds of traditional iterative reconstruction method, i.e., unfolding the iterations into sequential neural network layers. Notable examples of these unfolded traditional iterative methods include ISTA-Net [43], ADMM-CSNet [44], and AMP-Net [45], and many of the refined versions [46], [47]. However, these methods also need a large training dataset, and most of them are specifically designed for images. The training process is time-consuming. When applied to new applications, the generality is significantly inferior to that of traditional iterative-based methods, which limits their practical utility.

In summary, current most effective category of AI methods for sparse solution of under-determined linear systems was first

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proposed 9 years ago, but real-world applications still use traditional iteration methods that proposed 30 or even 50 years ago, such as OMP, ADMM, ISTA, IHT, which are time-consuming. The reason is pre-trained AI methods exhibit poor generality beyond their training conditions and dataset distributions, or lack interpretability. They require a model training process and a complex neural model that has a large quantity of trainable parameters. The model must be re-trained whenever there is a change in the measurement rate, measurement matrix, signal dimension, or application. Considering the need for model re-training, these methods are far from efficient. In many applications, it is challenging to collect a dataset that allows these training methods to achieve reasonable reconstruction quality. To advance CS reconstruction to a practical application level in the AI era, inspired by the idea of physics informed neural network [48], [49], [50], this paper proposes a training-free artificial neural method called CL that fully integrates the advantages of traditional iterative reconstruction and neural networks, which is able to truly replace traditional iterative methods.

III. Methods

A. Mathematic Model

The basic model of compressed sensing is designed for 1D signal $\mathbf{x} \in \mathbb{R}^{n\times 1}$ to get a measurement $\mathbf{y} \in \mathbb{R}^{m\times 1}$, $n \gg m$ so the measurements are compressed comparing to the original signal. So high-dimensional data is usually reshaped into 1D. The model is,

$$\mathbf{y} = \mathbf{M}\mathbf{x} + \xi = \mathbf{M}\mathbf{D}\mathbf{s} + \xi = \mathbf{A}\mathbf{s} + \xi \tag{1}$$

where $\mathbf{M} \in \mathbb{R}^{m\times n}$ and $\mathbf{A} \in \mathbb{R}^{m\times n} = \mathbf{M}\mathbf{D}$ are the sensing matrix and measurement matrix, respectively. $\mathbf{s} \in \mathbb{R}^{n\times 1}$ is the sparse coefficients, which is the sparse representation of $\mathbf{x}$ on the sparse basis $\mathbf{D} \in \mathbb{R}^{n\times n}$, ξ is the noise term. The reconstruction process is solving the following problem,

$$\min \|\mathbf{s}\|_0 \quad s.t. \|\mathbf{A}\mathbf{s} - \mathbf{y}\|_2 < \varepsilon \tag{2}$$

where $\|\cdot\|_0$ means the 0-norm. So CS reconstruction is use as less components in $\mathbf{s}$ as possible the make the residual $\mathbf{r} = \mathbf{A}\mathbf{s} - \mathbf{y}$ less than a threshold, ε. Methods like OMP, IHT, ISTA can be used to solve this problem without training. For a group of signals like image, the original signal $\mathbf{X} \in \mathbb{R}^{n\times a}$ and measurement results $\mathbf{Y} \in \mathbb{R}^{m\times a}$ are matrices, each column of $\mathbf{X}$ usually share the same measurement matrix as is shown in (3).

$$\mathbf{Y} = \mathbf{M}\mathbf{X} + \xi = \mathbf{M}\mathbf{D}\mathbf{S} + \xi = \mathbf{A}\mathbf{S} + \xi \tag{3}$$

This paper introduces a new reconstruction model as presented in (4), where $\mathbf{S}_i$ represents the i-th column in $\mathbf{S}$. 1D signal is a special case of this model where $a = 1$. In contast to traditional reconstruction problems, prior information is integrated into this model. This information is application-dependent and may include known features of $\mathbf{x}$ or $\mathbf{s}$. For example, when Discrete Cosine Transform (DCT) is used as the sparse basis, prior knowledge for 1D signals can be applied. If the data have a limited frequency range, a mask can be added to the signal $\mathbf{S}$ to be reconstructed. In another case, some multi-channel signals exhibit group sparsity, meaning their non-zero coefficients appear at the same positions. Group structure constraints such as L_2 or L_1 norms can then be used. For prior information on 2D signals, some images are relatively smooth, like thermal images. This implies that the high-frequency components in $\mathbf{S}$ are small, so a high-frequency constraint can be imposed. In other cases, such as certain medical images, different regions have well-defined numerical ranges, allowing spatial weighting constraints to be applied. This paper proposes two universal prior information, i.e., total variation (TV) and local variation (LV). TV is suitable for signals that contain few or no sharp edges or step changes, while LV applies to signals with minimal local variations. Most signals in the physical world adhere to this prior. Total variation for 1D signal and images are given in (5) and (6) respectively, where H and W are the pixel number in height and width of the image. TV suppresses outliers and ensures that the reconstructed signal is smooth.

$$\sum_i \min \|\mathbf{S}_i\|_0 \quad s.t. \begin{cases} \|\mathbf{A}\mathbf{S} - \mathbf{Y}\|_2 < \varepsilon \\ prioriinformation \end{cases} \tag{4}$$

$$TV(\mathbf{x}) = \frac{1}{n}\sum_{j=1}^{n-1}(x_j - x_{j+1})^2 \tag{5}$$

$$TV(I) = \frac{1}{H}\sum_{i=1}^{H-1}(I_{i,j} - I_{i+1,j})^2 + \frac{1}{W}\sum_{j=1}^{W-1}(I_{i,j} - I_{i,j+1})^2 \tag{6}$$

The local variation is for images and high-dimensional signals, the motivation is a local region is smooth, such as a small block in images. LV is defined in (7) as,

$$LV(I) = \frac{1}{B}\sum_{b=1}^{B} std(I_b) \tag{7}$$

where B is the segmented block numbers for the high-dimensional signal, $std(\cdot)$ is the standard deviation. For ease of implementation, a fixed-size sliding window can be used for block segmentation. The sliding step-size should smaller than the window size to prevent block effects in the reconstruction results. The default window size is 3×3 to balance between image details and local smooth.

B. The Proposed Ultra Small Model — Coefficients Learning

1) General Idea of Coefficients Learning: Many iterative reconstruction algorithms can be divided into two blocks, the first block injects some new information from the residual, while the second block forward current $\mathbf{s}$ and new residual information to update the estimation. The residual injection block typically utilizes the correlation between the residual and each column of $\mathbf{A}$ to get new information from the residual, i.e., $\mathbf{A}^T\mathbf{r}$, such as OMP, IHT, ISTA, subspace pursuit (SP), CoSaMP, etc. Instead of calculating the residual by $\mathbf{r} = \mathbf{A}\mathbf{s} - \mathbf{y}$, some methods uses $\mathbf{y}$ and some adjustable parameters. For example, iteratively reweighted least squares (IRLS) computes the residual as $\mathbf{W}\mathbf{y}$, where $\mathbf{W}$ is a weighting matrix that is adjusted in each iteration. No matter how the residual is calculated, since $\mathbf{A}$ is fixed, $\mathbf{A}^T\mathbf{r}$ can be implemented with artificial neural network structure, as

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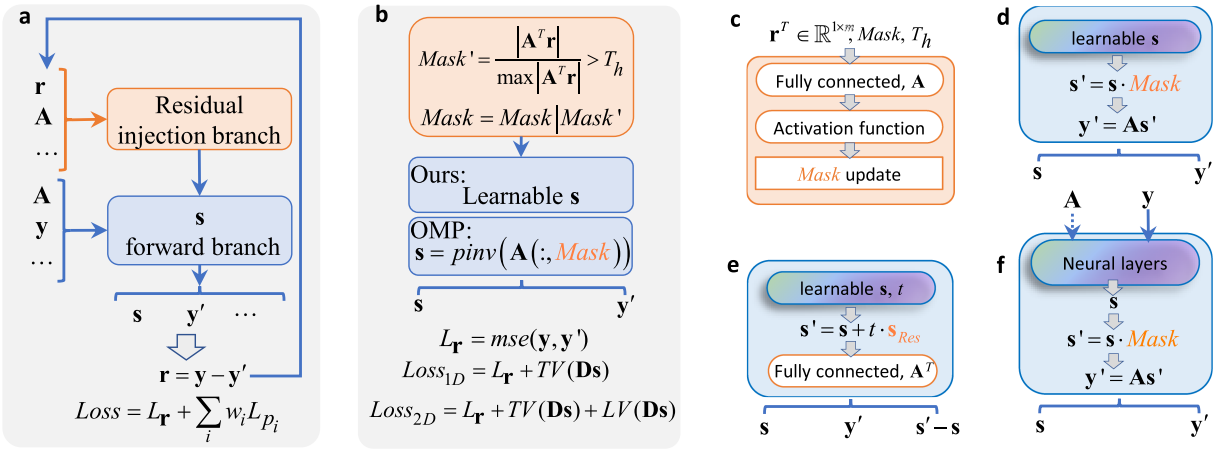


Fig. 1. Diagram of the proposed ultra small artificial neural models—coefficients learning (CL) for training-free and universal sparse reconstruction. (**a**) CL reforms traditional iterative methods into one residual injection branch and one $\mathbf{s}$ forward branch with learnable $\mathbf{s}$, so $\mathbf{s}$ is learned with the residual and prior loss, thereby eliminating the need for a large-scale training dataset while maintaining generality and interpretability. (**b**) shows the block function for CLOMP and general prior loss for 1D and 2D signals, where the learnable s is used to substitute the psduo-inverse operation. More implementation structures for the two branches are provided in (**c** – **f**). Since $\mathbf{s} \in \mathbb{R}^{n\times 1}$, the structures in (**c** – **e**) only need a minimal of n trainable parameters. These structures support parallel processing, so CL works for both 1D and high-dimensional data.

TABLE I
$\mathbf{s}$ FORWARD BLOCK FOR SOME SPARSE RECONSTRUCTION METHODS

Algorithms	$\mathbf{s}$ forward	Note
OMP, MP, CoSaMP	$\mathbf{s} = pinv\left(\mathbf{A}\left(:, Mask\right)\right)$	$Mask$ is a binary sequence derived from $\mathbf{A}^T\mathbf{r}$ to indicate the sub-columns
IHT	$\mathbf{s} = H_s\left(\mathbf{s} + t \cdot \mathbf{A}^T\mathbf{r}\right)$	H_s is a thredholding function to select the largest values, t is a scale factor
ISTA	$\mathbf{s} = soft_\lambda\left(\mathbf{s} - 2t \cdot \mathbf{A}^T\mathbf{r}\right)$	$soft_\lambda$ is a soft threshold function, t is a scale factor
IRLS	$\mathbf{s} = inv\left(\mathbf{A}^T\mathbf{W}\mathbf{A} + \lambda\mathbf{I}\right)\mathbf{A}^T\mathbf{W}\mathbf{y}$	$\mathbf{W}$ is an adjustable weighting matrix, λ is a scale factor

shown in Fig. 1(c), where $\mathbf{A}$ or $\mathbf{A}^T$ becomes a fixed parameter of a fully connection layer (or other structure like convolutional layer). Additional operations for $\mathbf{A}^T\mathbf{r}$ can be expressed as activation functions following the fully connected layer. Therefore, the first core idea of coefficients learning is reforming the residual injection into an artificial neural network structure with fixed parameters for $\mathbf{A}$ or $\mathbf{A}^T$ to expedite the calculation, which is called residual injection branch, as illustrated in Fig. 1(a).

The $\mathbf{s}$ forward block relies on different methods, some popular methods are listed in Table I, where $pinv(\cdot)$ and $inv(\cdot)$ denote the pseudo-inverse and inverse functions, respectively. These methods typically involve operations with high computational complexity, such as matrix inversion, pseudo-inversion, and threshold functions, which can be quite intricate. Therefore, the second core idea of coefficients learning is to make $\mathbf{s}$ becomes learnable parameters, rather than relying on the closed-form solution in Table I, which is called $\mathbf{s}$ forward branch. $\mathbf{s}$ are automatically learned with the help of residual injection branch and the loss function, i.e., the residual loss $L_\mathbf{r}$ and the prior information loss $\sum_i w_i L_{p_i}$, where w_i are scale factors for different prior terms.

The general framework of coefficients learning is summarized in Fig. 1(a), where the residual injection branch takes inputs such as $\mathbf{r}$, $\mathbf{A}$, or other parameters like $\mathbf{W}\mathbf{y}$, and produces information to help $\mathbf{s}$ forward. The $\mathbf{s}$ forward branch can also takes these parameters as input or operate without any input. The outputs are at least $\mathbf{s}$ and the estimation of $\mathbf{y}' = \mathbf{A}\mathbf{s}$. Some implementation structures for $\mathbf{s}$ forward branch are illustrated in Fig. 1(d)–(f). In Fig. 1(d), (e), $\mathbf{s}$ is directly set as learnable parameters, another option is Fig. 1(f), where a neural block can be employed that takes fixed input such as $\mathbf{y}$ with $\mathbf{A}$, and utilizes various neural layers to output $\mathbf{s}$. To implement the 0-norm in (4), the residual injection branch outputs a $Mask$ that indicates which positions are involved in the calculations for support-based methods such as OMP, SP, CoSaMP, and others. This mask is multiplied with $\mathbf{s}$ before being linked to the subsequent layers. For addition-based methods like IHT and ISTA, as illustrated in Fig. 1(e), the learnable $\mathbf{s}$ can be updated by minimizing $\mathbf{s}' - \mathbf{s}$. Algorithms like IRLS do not require the sparse positions of $\mathbf{s}$, in this case, both $\mathbf{s}$ and $\mathbf{W}$ can be set as learnable parameters to supervise each other mutually. During coefficients learning, any prior information can be incorporated into the loss function, resulting in improved accuracy compared to the baseline algorithm (e.g., CLOMP versus OMP).

2) Coefficients Learning on OMP (CLOMP): Taking OMP as an example, this section demonstrates how OMP is adapted into the proposed CL framework, thereby bypassing the complex pseudo-inverse operation. OMP iteratively identifies additional

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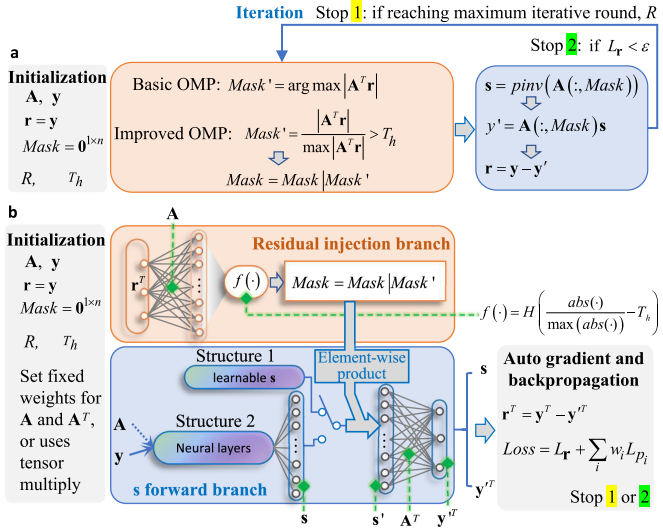


Fig. 2. Structure of the proposed coefficients learning on OMP (CLOMP). The process for OMP is shown in (**a**). (**b**) shows the structure for the residual injection branch and **s** forward branch for CLOMP, where $H(x)$ is the unit step function which is 0 for $x > 0$, otherwise 1.

columns in $\mathbf{A}$ that exhibit a high correlation with the residual. Subsequently, the coefficients are computed using the pseudo-inverse of all the highly correlated sub-columns of $\mathbf{A}$. The positions of these sub-columns correspond to the locations of the non-zero coefficients in $\mathbf{s}$. So, it works in one greedy manner and is efficient when the signal is sufficiently sparse, as the pseudo-inverse is applied only to the sub-columns.

As shown in Fig. 2(a) for OMP, the residual injection block calculate $\mathbf{A}^T\mathbf{r}$ first in each iteration. This process can be mapped into an artificial neural network structure, as illustrated in the residual injection branch in Fig. 2(b) and (8), where $\mathbf{W_A}^{(m,n)}$ denotes a bias-free fully-connected layer with m input neurons and n output neurons, whose weights are fixed as $\mathbf{A}$, and this can be set during the model initialization stage. $f(\cdot)$ represents the activation function as illustrated in Fig. 2(b).

$$Mask' = \left\{ i : \frac{|\mathbf{A}^T\mathbf{r}|_i}{\max|\mathbf{A}^T\mathbf{r}|_i} > T_h \right\} \mapsto f(\mathbf{r}^T\mathbf{W_A}^{(m,n)}) \quad (8)$$

The basic OMP only identifies the maximum positions in $\mathbf{A}^T\mathbf{r}$, which is relatively slow. The improved OMP, sometimes referred to as stOMP in the literature, identifies multiple positions in a single iteration instead. After passing $\mathbf{r}^T$ through this layer, $\mathbf{A}^T\mathbf{r}$ is computed. Next, the absolute operation and normalization process can be expressed as an activation function $f(\cdot)$, or using $softmax$ directly after taking the absolute value. The mask update is performed using a bitwise or operation between the current and previous mask, which is also very fast. For the $\mathbf{s}$ forward branch, the two structures depicted in Fig. 2(b) can be used. We find that structure 1 is more efficient than structure 2 with fixed and less trainable parameters. Therefore, the results section employs structure 1. Structure 2 has more learnable parameters, which may enhance accuracy in applications with more prior information. Next, $\mathbf{s}$ is multiplied by the mask in an element-wise manner to implement the 0-norm. Subsequently, another fully connected layer that uses $\mathbf{A}^T$ as fixed parameters can be employed to replace the $\mathbf{y}' = \mathbf{A}(:, Mask)\mathbf{s}$, as is shown in (9),

$$\mathbf{y}' = \mathbf{A}(:, Mask)\mathbf{s} \mapsto \mathbf{y}'^T = Mask \odot \mathbf{s}\mathbf{W_A}^{(n,m)} \quad (9)$$

where $\odot$ is element-wise product. Finally, the loss function in Fig. 1(b) can be applied to 1D and 2D signals. To replace the time-consuming $pinv(\cdot)$ in OMP, (10) is used.

$$\mathbf{s} = pinv(\mathbf{A}(:, Mask)) \mapsto \mathbf{s} = \mathbf{s} - \eta\nabla Loss \quad (10)$$

where η is the learning rate. With the auto-gradient to get the gradient ∇ and backpropagation capabilities of neural networks, $\mathbf{s}$ is learned automatically.

This CLOMP example illustrates why CL methods possess the same generality and interpretability as traditional iterative methods. In terms of generality, first, CL methods accept the same input as traditional iterative methods and output $\mathbf{s}$ or $\mathbf{x}$. Sampling matrices are fixed as neural network layers during model initialization, while trainable coefficients $\mathbf{s}$ of varying lengths are also initialized. Therefore, CL methods are applicable to any sampling matrix, sampling rate, and data size without requiring modifications. Second, $\mathbf{s}$ is automatically learned through residuals during the iterative process, eliminating the need for pretraining or data collection, achieving a out-of-the-box, nomodification workflow. Regarding interpretability, the decision process of CL methods is exceptionally clear. It inherits the traditional iterative approach of progressively optimizing sparse coefficients via residuals, the constraint terms, and the property that the output directly provides the sparse coefficients $\mathbf{s}$. Then, similar to traditional iterative methods, the original signal $\mathbf{x}$ is obtained by expanding the output on a sparse basis. The mechanism of how $\mathbf{s}$ influences $\mathbf{x}$ is welldefined and controllable. For example, in the DCT domain, each coefficient in $\mathbf{s}$ corresponds to a specific frequency component of the signal. Thus, the entire process of recovering the original signal is completely consistent with traditional iterative methods, with no opaque systems steps, fully preserving their interpretability. In contrast, other AI methods that directly target $\mathbf{x}$ through opaque systems training, such as Deep Unfolding, may claim interpretability in their forward process or architecture, but they remain fundamentally datadistributiondependent opaque systems models, suffering from rapid performance degradation on different data distributions or sizes. In terms of accuracy, CL methods can incorporate prior information into the loss function, a capability surpassing traditional iterative algorithms, leading to significantly improved accuracy. Regarding efficiency, if a traditional iterative method involves computationally expensive steps, CL can substantially enhance efficiency. For example, the absence of a pseudo-inverse design contribute to CLOMP's speed significantly, particularly when handling large-scale data.

IV. Experiments and Results

A. One-Dimensional Signal

1) Evaluation With Synthetic 1D Data: The fundamental model of compressed sensing is $\mathbf{y} = \mathbf{M}\mathbf{x} + \xi = \mathbf{M}\mathbf{D}\mathbf{s} + \xi =$

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$\mathbf{As}+\xi$. $m=\lfloor nSr \rfloor$, where Sr is the sampling rate. $\mathbf{s}\in\mathbb{R}^{n\times 1}$ is the sparse coefficients, which has K non-zeros values, and the sparse rate is defined as $Kr=K/n$. $\mathbf{s}$ links to the original signal $\mathbf{x}$ and sparse basis $\mathbf{D}\in\mathbb{R}^{n\times n}$ as $\mathbf{x}=\mathbf{Ds}$, ξ is the noise term. So the key factors influencing the reconstruction accuracy of CS are n, Sr and Kr. To validate the proposed method, this section generates $\mathbf{x}$ and tests it under various combinations of (n, Sr, Kr) within their typical ranges, with 1000 random signals generated for each combination. For practical applications involving 1D data, segmented sensing is often employed. The segment length rarely exceeds 10^4, while lengths below 100 tend to cause high inter-segment variation under low sampling intervals, thereby reducing sparsity. Hence, the typical range for n is from 10^2 to 10^4. A lower Kr indicates better sparse representation and easier reconstruction. In practice, Kr values seldom fall below 0.01 or exceed 0.8. Other CS studies, such as ISTANetand AMPNet, generally test Kr values between 0.01 and 0.5. Accordingly, this paper sets the Kr range from 0.01 to 0.8. Regarding Sr, some traditional iterative methods have theoretical lower bound for minimal recoverable Sr. AI-driven approaches, through data-driven end-to-end learning, can automatically leverage additional information to further push below the bound. When Sr exceeds 0.5, the reduction in sampling time and data volume no longer justifies the increased reconstruction overhead. Since existing AI methods lack comparable generality, we compare our approach with traditional iterative methods in this experiment. Therefore, Sr is set from 0.05 to 0.5, which is considered recoverable. Experiments are then conducted comparing OMP, CLOMP, and another typical greedy algorithm known for its speed with a known Kr, IHT. Unless stated otherwise, the normalized cut-off threshold (T_h) in OMP and CLOMP for selecting support in each iteration and maximum iteration number (R) for all methods, are set to 0.7 and 200 in this paper, respectively. The results are presented in Figs. 3, 4, and 5.

To generate $\mathbf{x}$ under the combination of (n, Kr), we initialized a length n signal with zeros and randomly selected K positions as the non-zero coefficients. DCT matrix is chosen as $\mathbf{D}$ (same for the rest of this paper), so $\mathbf{s}$ is the frequency components of the signal. A real-world signal typically exhibits dominant values in the low-frequency range and much smaller values in the high-frequency range. To ensure the signals are representative, the sorted non-zero coefficients follow the shape of a Gaussian probability density function, where the low-frequency part contains a small portion of dominant values, accompanied by a long tail distributed in the rest of $\mathbf{s}$. ξ is considered as small random values in the high-frequency part. An example for ($n=10^3$, $Kr=0.2$) is illustrated in Fig. 3(a). $\mathbf{M}$ is set as random Gaussian matrix (the same notation will be used throughout this paper). To evaluate the similarity between the reconstruction and the ground truth, we utilize several metrics: the Structural Similarity Index Measure (SSIM), Peak Signal-to-Noise Ratio (PSNR), Mean Squared Error (MSE), and Pearson Correlation Coefficient (PCC). SSIM is selected as the primary index because it effectively combines both local and global information. These metrics are calculated for each signal

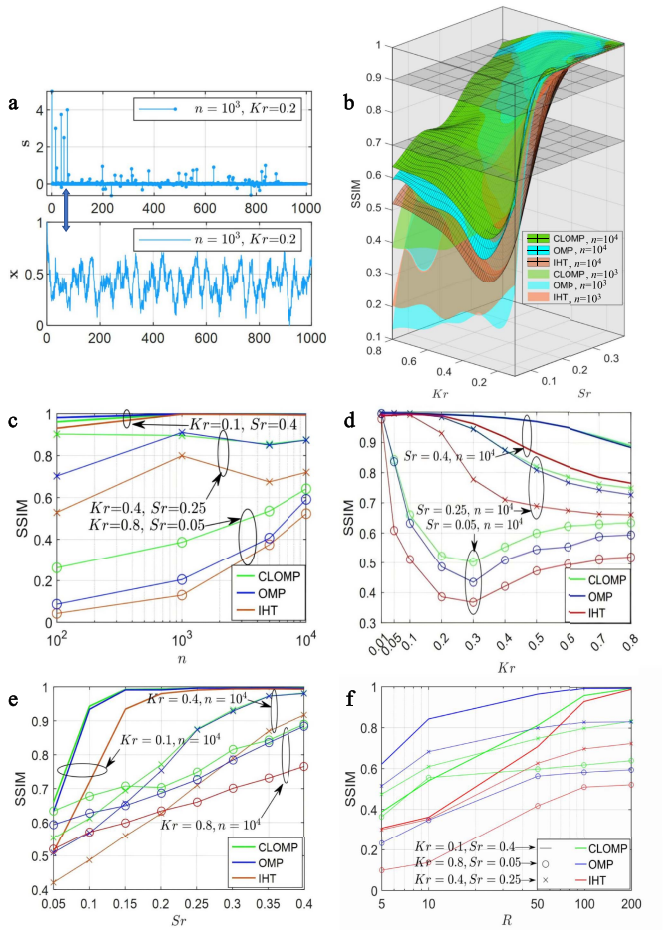


Fig. 3. Reconstruction accuracy based on SSIM in relation to the sparse rate (Kr), sampling rate (Sr), signal length (n) and iteration number (R) for 1D signals. ($\mathbf{a}$) is an example of a synthetic 1D signal with $n=10^3$ and $Kr=0.2$, where the coefficients $\mathbf{s}$ and the time-domain signal $\mathbf{x}$ are presented. The SSIM values for a demo method of the proposed framework (CLOMP) and its mother method OMP, and IHT are presented in ($\mathbf{b}-\mathbf{e}$). ($\mathbf{c}-\mathbf{f}$) shows the typical cases with the combination of Kr, Sr and n for SSIM versus n, Kr, Sr and R, respectively.

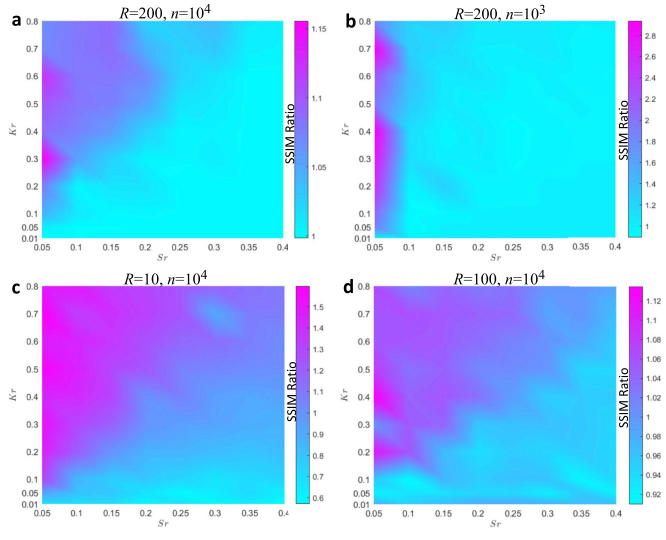


Fig. 4. The reconstruction accuracy ratio of CLOMP to OMP based on SSIM in relation to Kr and Sr under ($\mathbf{a}$) $n=10^4$ and ($\mathbf{b}$) $n=10^3$ for 1D signals.

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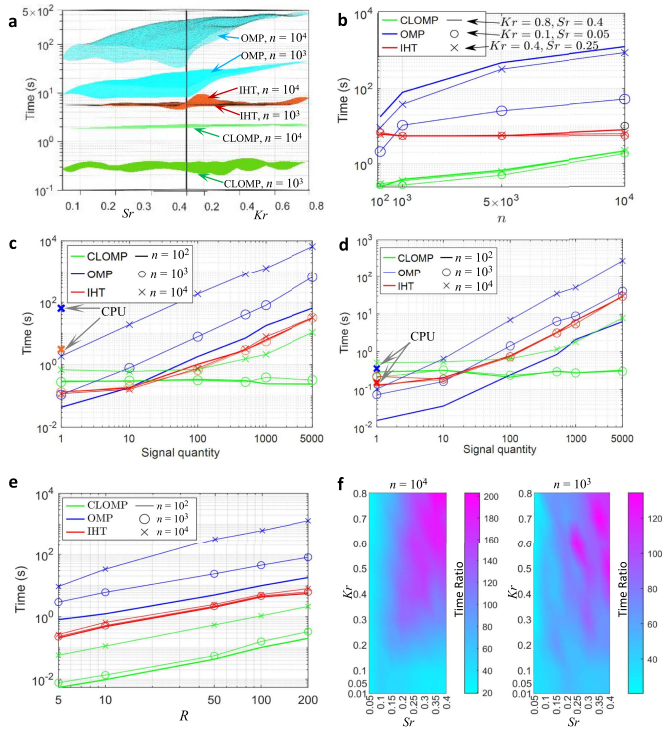


Fig. 5. The reconstruction time of the proposed CLOMP and other methods versus Sr, Kr, n, R and signal quantity for 1D signals. (**a**) is a 3D plot for time versus Sr and Kr under different n. Time versus n for typical best, middle, and worst cases are denoted with 'o', '×', and line in (**b**), respectively. (**c**, **d**, **e**) show the time versus signal quantity or R under worst and best cases for the combination of Sr, Kr, respectively. (**f**) is the time ratio of CLOMP to OMP. The implementation of OMP and IHT under $n = 10^4$ on CPU is also highlighted for the case where the signal quantity is 1.

individually before obtaining the mean value for the 1000 signals under each combination. The algorithms are implemented using PaddlePaddle on a computer with an Intel Xeon Platinum 8352Y CPU and an Nvidia RTX4090 GPU. Since traditional iterative algorithms often involve substantial logical operations, they may execute faster on CPUs than on GPUs for small-scale data. The processing times for both GPU and CPU implementations are provided primarily to benchmark the proposed method against the fastest available runtime of each baseline, whether on GPU or CPU. This comparison is not intended to highlight the parallel or multi-core support of the proposed method itself.

Figs. 3(b)–(e) and 4 demonstrates that the proposed method exhibits significantly improved reconstruction quality in low Sr and high Kr regions, achieving approximately 2 to 3 times of SSIM compared to OMP, while maintaining similar accuracy to OMP in other regions. Lower values of Kr (i.e., better sparisity condition) generally yield better performance across most sampling rates for all methods, e.g., when $n = 100$ and $Kr = 0.01$, only one significant position needs to be identified from 100 positions. As a result, the performance of all methods approaches near-perfect recovery. Higher values of Sr consistently lead to enhanced performance, as they allow for the extraction of more information from the original signal. A high Kr may result from inadequate sparse representation, signal complexity, or measurement noise, which requires higher Sr to achieve satisfactory performance. The proposed method performs better in these challenging scenarios. Fig. 3(c) demonstrates that a longer n has better performance in most cases, because $\mathbf{A}$ has more detailed basis and a lower average residual threshold per signal point. The reason why the proposed method has better accuracy is that estimation error in $\mathbf{s}$ can be corrected with the loss backpropagation. If OMP gets some wrong support positions in $\mathbf{s}$, these erroneous positions will persist throughout the remaining iterations, leading to error accumulation and progressively deteriorating position estimation. This advantage becomes more significant in low Sr where wrong position in $\mathbf{s}$ are more likely to be obtained during the early iterations. Fig. 3(f) illustrates the effect of iteration count on accuracy. Under challenging conditions with low sparsity and low sampling rates, the CLOMP method achieves performance close to the final result of OMP in as few as 10 iterations. Under moderate or favorable conditions, CLOMP approaches the performance of OMP within 100 to 200 iterations.

In terms of efficiency in Fig. 5, on the synthetic dataset with 1000 signals, the proposed method saves 2∼3 orders of magnitude in time consumption than OMP and 1∼2 orders of magnitude than IHT generally, as illustrated in Fig. 5(a)–(f). The time of proposed method is not sensitive to Sr, Kr, n and signal quantity, so the efficiency improvement particularly pronounced in challenging scenarios where these parameters are high. For instance, when $Kr = 0.01$ and $Sr = 0.05$, the efficiency advantage of CLOMP becomes less pronounced. Fig. 5(e) shows the processing time of all method increases almost linearly with R. Therefore, R does not affect their relative time performance. The traditional OMP and IHT exhibit greater efficiency only when recovering a limited number of signals with $n <$1000, as shown in Fig. 5(c)–(d), the proposed method maintains low time consumption in this context while achieving superior accuracy. With the increase of n and signal quantity, the proposed method surpasses OMP in efficiency gradually, which demonstrates the efficiency of CLOMP to OMP, especially on large-scale data. Example applications for single or several signal with large n are real-time high-resolution sensing [51], point-to-point wireless data transfer, recovering high-dimensional signals by reshaping them into 1D shape, etc. Example applications for large signal quantity are wireless sensor networks [52], image and other high-dimensional data reconstruction, multi-pass data transfer, and parallel reconstructions.

2) Test on Real 1D Signals: 1D signal datasets with varying average sparse rates are examined in this study. The Kr is calculated by descending sort the absolute values of the frequency components first, and then accumulating the sorted result, if a accumulated value reaches 0.98 of the total summation, Kr is the ratio of the position index to n. Average sparse rates are obtained with the average Kr of each signal within a dataset. We selected three datasets from electrocardiogram (ECG), electroencephalography (EEG), and wireless sensor networks (WSN), because compressed sensing has significant potential in these applications for conserving power or compress data of sensor

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TABLE II
TEST RESULTS ON REAL 1D DATASETS. THE MEAN SPARSE RATE IS DENOTED AS $\bar{K}r$. THE IMPROVEMENT ARE CALCULATED BY COMPARING THE PROPOSED CLOMP AND BEST ONE AMONG OMP AND IHT.

Dataset	Sr	Time (s)	Mean SSIM	Mean MSE	Mean PCC	Mean PSNR
Type $\bar{K}r$		CLOMP, OMP-CPU/GPU, IHT-CPU/GPU	CLOMP/OMP/IHT, Improvement (%)			
ECG	0.05	0.227, 6.986/26.661, 20.753/6.716	0.549/0.344/0.417, 31.65	0.026/0.055/0.039, -33.3	0.519/0.222/0.231, 124.68	16.239/13.556/14.435, 12.50
0.549	0.25	0.314, 164.246/136.008, 25.791/6.256	0.903/0.628/0.657, 37.44	0.003/0.015/0.015, -80.0	0.955/0.774/0.738, 23.39	26.379/19.201/18.846, 37.38
	0.5	0.319, 344.126/197.088, 35.563/6.157	0.991/0.970/0.847, 2.16	0.0002/0.0006/0.004, -66.67	0.997/0.989/0.927, 0.81	37.622/33.359/24.311, 12.78
WSN	0.05	0.319, 0.814/0.591, 0.990/0.191	0.704/0.476/0.427, 47.90	0.023/0.022/0.031, 4.55	0.363/0.356/0.284, 1.97	16.918/16.807/15.404, 0.66
0.897	0.25	0.312, 23.214/6.188, 3.554/0.170	0.812/0.613/0.604, 32.46	0.006/0.012/0.015, -50.0	0.811/0.664/0.583, 22.14	22.669/19.522/18.545, 16.12
	0.5	0.327, 100.919/19.082, 13.136/0.149	0.886/0.790/0.759, 16.73	0.003/0.005/0.007, -40.0	0.917/0.860/0.797, 6.63	26.611/23.898/22.278, 11.35
EEG	0.05	0.476, 3.705/28.882, 17.057/7.956	0.735/0.254/0.404, 81.93	0.029/0.099/0.066, -56.06	0.614/0.275/0.280, 119.29	15.886/10.985/12.250, 29.68
0.751	0.25	0.276, 66.716/93.758, 24.791/8.056	0.921/0.712/0.645, 29.35	0.004/0.013/0.017, -69.23	0.955/0.839/0.722, 13.83	24.799/20.080/18.779, 23.50
	0.5	0.269, 133.686/137.207, 25.864/8.073	0.980/0.977/0.865, 0.31	0.001/0.001/0.005, -10.34	0.992/0.991/0.923, 0.10	31.590/31.193/24.250, 1.27

greatly. More specifically, implantable biomedical devices like cardiac pacemaker are limited in power and difficult to charge. These devices usually equip with low-power sensors to capture signals like ECG and EEG. With CS, these implantable sensors only need to wake up for a brief period during each cycle, resulting in substantial power savings. Similar advantages are also appealing for WSN. So, this paper selects Physionet2017ECG,[1] Slow Cortical Potentials,[2] and AirQuality[3] for ECG, EEG, and WSN, their signal lengths are shown in Fig. 6(j), their signal quantity are 8529, 2660, and 12, respectively. To save time, if the signal quantity is more than 1000, this study only takes the first 1000 signals. The test results are shown in Fig. 6 and Table II.

A piece of reconstructed signal with a length of 100 from the three datasets is shown in Fig. 6(a)–(i). The ECG signal is smooth, exhibiting a good Kr of 0.324, the WSN signal is more fluctuating, with a Kr of 0.846, making it more challenging to reconstruct. The EEG signal falls between the two. These sub-figures show SSIM is more strict to define perfect reconstruction than PCC and consider both local and global similarity well. This is why SSIM is chosen as the primary metric. For instance, the OMP reconstruction near the signal index of 50∼60 has obvious local difference from the ground truth, PCC gets a value of 0.995 while the proposed CLOMP gets a lower value of 0.978. At a high sampling rate of 0.5, only the proposed method achieves an SSIM greater than 0.9 of SSIM on all signals, while IHT exhibits the worst performance. Under a low Sr of 0.05, the performance improvement of the proposed method is particularly pronounced. Fig. 6(i) shows OMP and IHT almost failed, but the proposed method still gets the overall shape of the ground truth. Due to better accuracy, the proposed method requires a lower Sr to reach SSIM $>$ 0.9. In resource-limited applications like WSN, lower Sr means more saving in data and power consumption. In medical sensing applications, a lower Sr results in reduced exposure to harmful X-rays for patients. For the three demo signals, the proposed method reduces Sr by 0.07, 0.31, 0.23, corresponding to -23.3%, -53.4%, -67.6% of reduction in measurement rates.

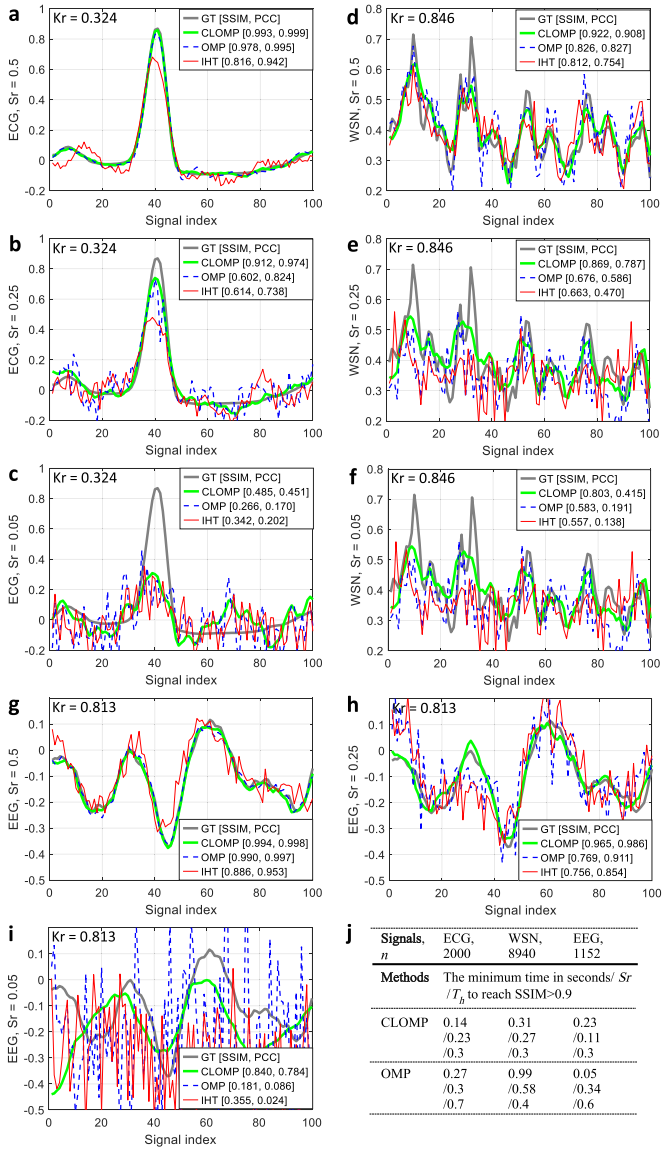


Fig. 6. Examples of reconstructions with a length of 100 on real 1D datasets. The ground truths are denoted as GT in ($\mathbf{a}-\mathbf{i}$). The sparse rates (Kr) of the full signal length (n) are given in the figure. SSIM and PCC of the demo signal are calculated and presented in the square brackets. ($\mathbf{j}$) provides further evaluation for the proposed CLOMP and its mother method OMP, where the minimum time and Sr to reach SSIM $>$ 0.9 are tested for the demo signal in ($\mathbf{a}-\mathbf{i}$), T_h and maximum iteration number are freely set for each method.

[1]https://www.kaggle.com/datasets/luigisaetta/physionet2017ecg
[2]https://www.kaggle.com/datasets/towsifahamed/eeg-dataset-of-slow-cortical-potentials
[3]https://www.kaggle.com/datasets/fedesoriano/air-quality-data-set

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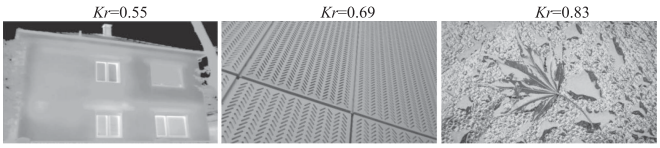


Fig. 7. A group of source images for synthetic 2D signals.

Table II presents the average performance on these datasets. The proposed CLOMP demonstrates significantly lower time consumption compared to the original OMP on all datasets. IHT is the most efficient on WSN dataset, because the time required for the GPU implementation of IHT is primarily influenced by the quantity of signals, as illustrated in Fig. 5, while WSN dataset only 12 signals. In most cases, the time of CPU implementation is much longer than it's GPU implementation, except for OMP in the EEG dataset and the low Sr regions for ECG dataset, because the signal length or average Kr for these two exceptions is relatively small, resulting in the time for tensor generation and data transfer exceeding that of parallel computing. Regarding accuracy, the proposed method consistently outperforms others across all metrics, with significant improvements noted in low Sr cases. For example, the proposed CLOMP achieves approximately 32%, 48%, and 82% of improvement in SSIM under a Sr of 0.05 for the three datasets, respectively.

B. Two-Dimensional Signal

1) Evaluation With Synthetic 2D Data: One popular method for reconstructing images and high-dimensional signals is reshaping the image into 1D signal, but the signal length will be excessively large. For example, one 128×128 image generates a signal length of $n = 128 \times 128 = 16384$, the $\mathbf{D}$ and reconstruction time increase significantly with higher image resolution. Another way is regarding them as a large collection of 1D signals, where each signal has a different $\mathbf{s}$ and corresponds to different columns in $\mathbf{A}$, so the pseudo-inverse operation for traditional reconstruction algorithm like OMP need to prepare different input for each row or column, which is slow. The proposed method makes $\mathbf{s}$ become learnable parameters, thereby bypassing the need for the pseudo-inverse operation.

To evaluate the performance on 2D data under important CS settings like Sr, Kr, n, R and signal quantity, this section utilizes three 4K resolution images with low, middle and high Kr, and resizes them to eight common resolutions ranging from 320×180 to 7680×4320. Since higher-resolution images can be processed in patches, testing is only conducted up to 8K resolution. In these images, n and signal quantity are the row and column numbers of an image, i.e., the resolution. A group of images is presented in Fig. 7. We find smaller resolution for the same image usually has higher Kr, to make the resized images keep the same Kr, Gaussian smoothing with varying parameters is applied. The parameter Sr is set within its typical range of 0.1 to 0.5. Grayscale images are employed for this analysis. In this test, local variation and total variation are utilized as prior information. The results are summarized in Figs. 8, 9, and 10.

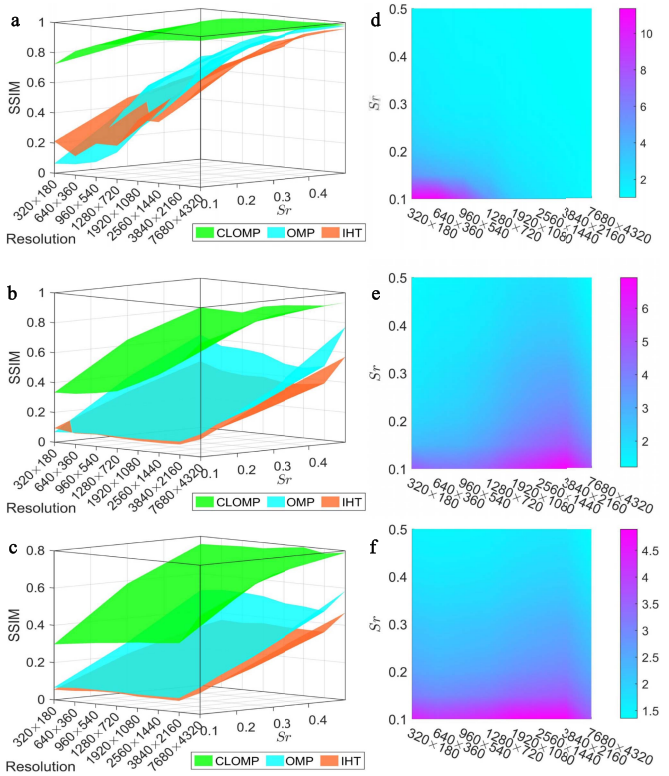


Fig. 8. The reconstruction accuracy based on SSIM in relation to image resolution, Sr and Kr for 2D signals. ($\mathbf{a}-\mathbf{c}$) are for the synthetic images with $Kr = 0.55, 0.69, 0.83$, respectively. ($\mathbf{d}-\mathbf{f}$) are the corresponding SSIM ratios of CLOMP to OMP for $Kr = 0.55, 0.69, 0.83$, respectively.

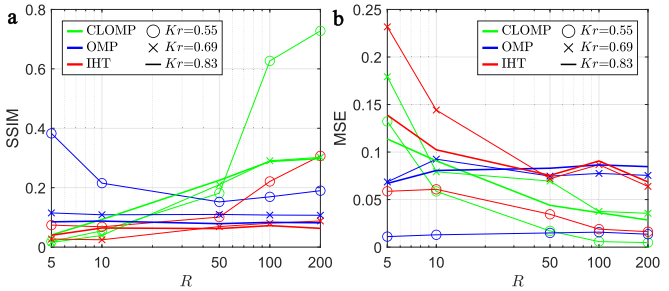


Fig. 9. The average reconstruction accuracy in relation to R for 2D signals based on ($\mathbf{a}$) SSIM and ($\mathbf{b}$) MSE.

The accuracy of the proposed method is at least twice as high as that of OMP in most cases, with improvements ranging from four to ten times under low Sr, as is shown in Fig. 8(d)–(f). Fig. 8(a)–(c) demonstrates that a lower sparsity rate generally leads to decreased accuracy. For instance, with an image resolution of 320×180 and $Sr = 0.1$, the three images get SSIM of 0.72, 0.35, 0.30 for their Kr of 0.55, 0.69, 0.83, respectively. With the increase of image resolution or Sr, the accuracy becomes better, and the influence of Sr is more significant than image resolution. This figure also indicates that the proposed method requires significantly less Sr to reach the same accuracy, which can greatly reduce the time during data acquisition, indicating great potential in medical imaging. Fig. 9

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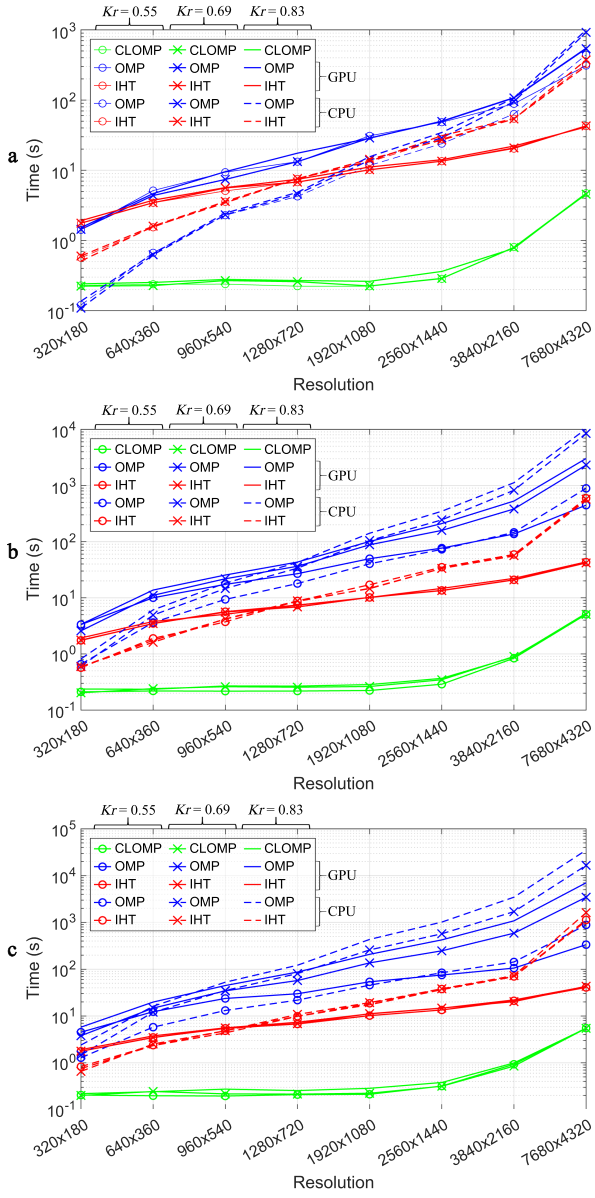


Fig. 10. Reconstruction time versus image resolution, Sr and Kr for 2D signals. ($\mathbf{a}-\mathbf{c}$) are for low, middle, and high sampling rates with $Sr = 0.1, 0.3, 0.5$, respectively.

indicates that CLOMP surpasses OMP after approximately 50 iterations. Beyond this point, the performance of OMP remains virtually unchanged, while CLOMP continues to improve until it plateaus around 200 iterations.

In terms of efficiency, Fig. 5(e) has demonstrated that the efficiency of all methods scales nearly linearly with R. Since the efficiency for a 2D signal is equivalent to that of a set of 1D signals with the same dimension, further testing is unnecessary. Fig. 10 indicates that other methods only possibly more efficient for super low resolution images that lower than 320×180, but a high-enough image resolution is important for almost all applications. Within the range of 320×180 to 7680×4320, the proposed CLOMP method is 2 to 3 orders of magnitude more efficient than traditional iterative greedy algorithms. For example, the proposed method can recover a 4K image in less than 1 second, while OMP needs at least approximately 100 seconds on GPU and around 1000 seconds on CPU for $Sr > 0.3$. The computational time for CLOMP only increases markedly at resolutions above 2560×1440, while other methods continue to be significantly affected by resolution across the tested range. This is because the matrix multiplication operation involving $\mathbf{D}$ and $\mathbf{s}$ in CLOMP becomes increasingly time-consuming at larger resolutions. Another observation is that images with larger Kr typically require more time to reconstruct, but the influence is not so significant as resolution, so a good sparse representation will enhance both accuracy and efficiency.

2) Test on Image Datasets With Various Resolutions, Modalities, and Sparse Rates: This section evaluates the proposed method on various image datasets and our collected datasets for CS with varying resolutions, modalities, and sparse rates. To maintain the same number of images as in SET11, we also selected 11 images from other dataset, which is sufficient to demonstrate the accuracy and efficiency of our method. We collected additional images with FHD, UHD, and FUHD resolution, so the image datasets covers the resolution from 320×240 to 7680×4320, the modalities include infrared, visible light, and MRI. The full dataset can be found here. We employed well-known metrics like SSIM, MSE, and PSNR. The detailed information like resolution and average sparse rates for the datasets is provided in Table III. One image from each dataset is displayed in Fig. 11. Note that we did not exhaustively optimize the proposed method during this test, adjusting the weight parameters may get better accuracy in these results. With a Sr of 0.1, the proposed method has 0.5, 7.5 of improvements on average on SSIM and PSNR for these images, respectively. At a Sr of 0.3, the average improvement still reach 0.33 and 5.87, respectively. Generally, a lower Kr leads to better accuracy. The zoom-in squares on the right column demonstrate that the proposed method effectively recovers both the abstract shape and fine details under $Sr = 0.3$ for all datasets. Since local variance is utilized as prior information in this study, images with a smooth local appearance exhibit better accuracy, such as the image selected from BSD68. Conversely, images with fluctuating local regions get lower accuracy with the proposed method, as seen in the fingerprint image selected from SET11. Both total variance and local variance are universal prior information for images that can be applied in most of scenes. In practical applications, users can based on these two priors and incorporate additional prior information into their loss function.

More comprehensive test results on these datasets are presented in Table III. Compared to OMP and IHT, the proposed method has 273%, 93%, 43% of improvement on average on SSIM for $Sr = 0.1, 0.3, 0.5$ respectively, and has 43%, 27%, 16% of improvement on average on PSNR. The MSE improvement are more consistent across all Sr with an average reduction of 67%. Thus, the proposed method significantly outperforms traditional iterative algorithms in accuracy. In terms of efficiency, the proposed CLOMP only 76 microseconds slower than the CPU implementation of OMP under $Sr = 0.1$ for the resolution of 320×240. It is much faster than other iterative algorithms for higher resolution images and higher Sr. On the 4K UHD dataset,

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TABLE III
TEST RESULTS ON DIVERSE IMAGE DATASETS. THE IMPROVEMENT ARE CALCULATED BY COMPARING THE PROPOSED CLOMP AND BEST ONE AMONG OMP AND IHT.

Datasets	**Sr**	**Time (s)**			**Mean SSIM**	**Mean MSE**	**Mean PSNR**
Name Resolution Mean Kr		CLOMP	OMP-CPU/GPU	IHT-CPU/GPU	CLOMP/OMP/IHT, Improvement(%)		
Facial IR	0.1	0.256	0.180/1.984	0.611/1.841	0.707/0.169/0.233, 203.43	0.009/0.065/0.043, -79.07	21.818/12.119/13.953, 56.37
320×240	0.3	0.227	0.992/4.304	0.702/1.850	0.844/0.498/0.478, 69.48	0.002/0.005/0.009, -60.00	28.663/23.252/20.797, 23.27
0.707	0.5	0.210	1.585/5.542	0.743/1.843	0.877/0.729/0.655, 20.30	0.001/0.001/0.003, -21.43	30.608/28.834/26.006, 6.15
BSD68	0.1	0.260	0.418/2.874	1.733/2.590	0.507/0.056/0.122, 315.57	0.015/0.075/0.050, -70.00	19.531/11.578/13.669, 42.89
481×321	0.3	0.258	3.073/7.667	1.883/2.609	0.665/0.293/0.268, 126.96	0.006/0.020/0.022, -70.00	23.956/18.113/17.592, 32.26
0.840	0.5	0.260	7.681/11.753	2.001/2.608	0.761/0.475/0.375, 60.21	0.003/0.009/0.012, -66.67	26.625/21.592/20.434, 23.31
SET11	0.1	0.264	0.920/3.853	2.740/2.487	0.471/0.045/0.085, 454.12	0.016/0.093/0.056, -71.43	18.525/10.403/12.829, 44.40
512×512	0.3	0.267	6.237/10.147	3.344/2.492	0.674/0.275/0.234, 145.09	0.006/0.022/0.024, -72.73	23.028/17.123/16.540, 34.49
0.821	0.5	0.265	16.884/16.558	3.790/2.487	0.777/0.490/0.373, 58.57	0.003/0.009/0.014, -66.67	25.627/21.104/19.017, 21.43
Urban100	0.1	0.266	3.831/12.840	8.438/5.590	0.465/0.054/0.100, 365.00	0.016/0.084/0.061, -73.77	18.744/10.996/12.992, 44.27
1024×768	0.3	0.272	29.515/34.193	11.489/5.558	0.651/0.282/0.237, 130.85	0.006/0.022/0.026, -72.73	23.008/17.592/16.807, 30.79
0.827	0.5	0.276	80.616/61.679	12.740/5.619	0.763/0.450/0.359, 69.56	0.003/0.011/0.015, -72.73	25.585/20.501/19.034, 24.80
Brain MRI	0.1	0.222	6.145/16.930	10.206/5.517	0.767/0.232/0.244, 214.34	0.003/0.025/0.029, -88.00	26.000/16.323/15.844, 59.28
1024×1024	0.3	0.223	35.376/39.706	13.135/5.534	0.853/0.570/0.462, 49.65	0.001/0.003/0.006, -66.67	33.368/26.258/22.471, 27.08
0.645	0.5	0.217	49.809/45.137	13.366/5.476	0.926/0.847/0.689, 9.33	0.001/0.001/0.002, -40.00	36.460/33.965/27.997, 7.35
FHD	0.1	0.263	13.746/32.857	21.950/10.548	0.536/0.085/0.139, 285.61	0.009/0.042/0.031, -70.97	21.727/14.138/15.770, 37.77
1920×1080	0.3	0.279	101.367/87.383	27.720/10.514	0.674/0.348/0.304, 93.68	0.004/0.013/0.015, -69.23	25.292/20.213/19.345, 25.13
0.844	0.5	0.268	291.367/159.825	27.498/10.471	0.764/0.499/0.420, 53.11	0.003/0.007/0.009, -57.14	27.329/23.235/21.785, 17.62
QHD	0.1	0.283	21.335/40.507	30.523/11.302	0.552/0.100/0.137, 302.92	0.007/0.033/0.029, -75.86	22.316/14.902/15.607, 42.99
2040×1356	0.3	0.316	167.056/117.493	32.142/11.289	0.705/0.339/0.294, 107.96	0.003/0.009/0.012, -66.67	26.420/20.909/19.770, 26.36
0.825	0.5	0.320	468.523/210.805	32.892/11.262	0.792/0.507/0.424, 56.21	0.002/0.005/0.007, -60.00	28.803/24.051/22.297, 19.76
4K UHD	0.1	0.718	84.276/107.045	54.510/20.857	0.707/0.219/0.234, 202.14	0.005/0.023/0.024, -78.26	25.813/17.980/17.970, 43.57
3840×2160	0.3	0.863	631.441/318.513	57.581/20.703	0.831/0.514/0.467, 61.67	0.002/0.007/0.008, -71.43	31.496/25.057/23.305, 25.70
0.777	0.5	0.915	1484.288/505.072	61.335/20.962	0.880/0.683/0.606, 28.84	0.001/0.003/0.004, -66.67	33.464/28.866/26.200, 15.93
8K FUHD	0.1	4.734	601.238/401.862	318.995/42.112	0.743/0.345/0.283, 115.60	0.003/0.009/0.013, -66.12	25.979/21.401/19.677, 21.40
7680×4320	0.3	5.736	4734.480/1412.983	515.897/42.218	0.848/0.539/0.481, 57.40	0.001/0.003/0.005, -71.11	31.915/26.426/24.263, 20.77
0.761	0.5	5.733	10863.652/2456.990	1199.872/43.306	0.878/0.694/ 0.599, 26.55	0.001/0.002/0.003, -63.75	33.865/30.368/27.287, 11.52

by substituting the pseudo-inverse with learnable parameters, CLOMP is 117 and 29 times faster than OMP and IHT with $Sr = 0.1$, respectively, and 552 and 23 times faster with $Sr = 0.5$. With the increase of image resolution, the time improvements become even more significant. MRI, infrared images and 4K images perform better than other scenes, because they have relatively lower sparse rate on DCT.

C. Generality Tests and Comprehensive Analysis

1) Generality Tests: To further assess the generality of the CL method, this section selects one algorithm from each of the three main categories of traditional iterative compressed sensing methods, specifically greedy algorithms, convex optimization, and message passing. These are MP, ISTA, and AMP, and implements CLbased versions of them. The performance of these implementations is then compared and analyzed against two classes of stateoftheart methods, namely conventional iterative methods and Deep Unfolding, using both real 1D and 2D datasets.

On 1D datasets, comparisons are made between CL and the corresponding traditional iterative methods, which share similar generality. Deep Unfolding models are excluded, because they are pretrained on images with fixed n and sampling matrices, requiring substantial modification and retraining, and thus are not directly applicable to 1D data of arbitrary length. The results are shown in Fig. 12.

As seen in Fig. 12(a)–(c), the CL methods outperform their respective baselines across all metrics. The most significant performance gain occurs within the Sr range of 0.04 to 0.25. For instance, on the EEG dataset at $Sr = 0.1$, CLMP achieves an SSIM improvement of 0.7 over MP, while CLISTA shows an improvement of approximately 0.42 at the same point. Within this range, traditional iterative methods have less reference data for reconstruction, whereas the CL methods incorporate more prior information, leading to rapid performance gains. Improvements are less pronounced in the extremely low sampling rate region near $Sr = 0.01$, as the available data is too scarce even for CL's prior information. In the region where $Sr > 0.4$, the reference data is sufficient for traditional iterative methods to achieve good results, leaving less room for improvement by CL methods. Fig. 12(d) shows that, except for CLISTA, both CLMP and CLAMP demonstrate markedly higher efficiency than MP and AMP. CLISTA involves additional loss term computation,

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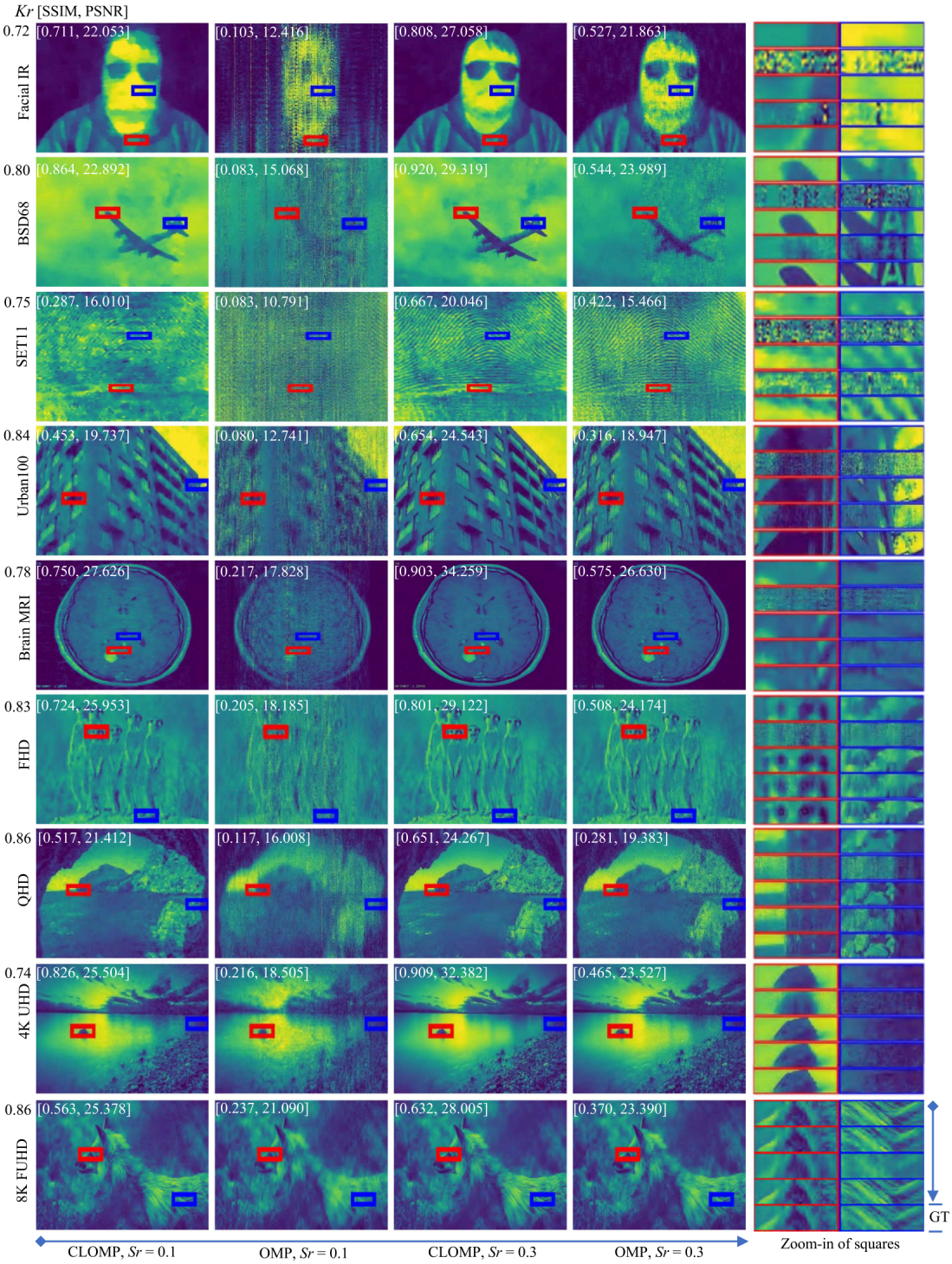


Fig. 11. Reconstruction examples. The zoom-in of squares are displayed in the rightmost column, where the bottom row is from the ground truth (GT) images. The image resolutions are shown in Table III.

and unlike ISTA, which lacks complex operations like matrix inversion. In the WSN dataset, which has the largest n and the poorest Kr, the computational cost of the **Ds** operation in the CL methods becomes increasingly noticeable as n grows.

On 2D image datasets, Deep Unfolding methods process images of different resolutions by dividing them into patches, which inherently includes operations like patch splitting and merging, making them directly applicable to images of varying resolutions. However, they are still trained on fixed sampling matrices. Therefore, this experiment compares the performance of the proposed CL methods, traditional iterative methods, and Deep Unfolding methods under two conditions, using their pre-trained fixed sampling matrices and using non-pre-trained random matrices. All methods were implemented in PyTorch

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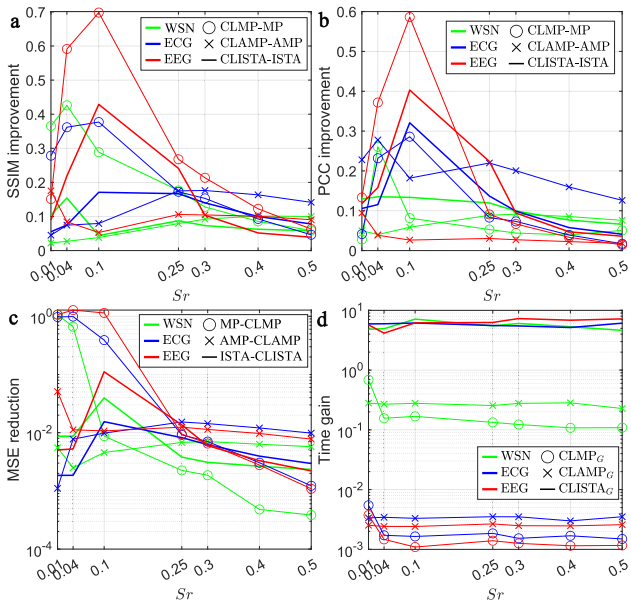


Fig. 12. The proposed CL methods versus their baseline methods. (**a**–**c**) show the difference in improvement amounts of the CL method under different quality metrics. (**d**) shows the time gain ratio of $CL/\hat{M}$, where $\hat{M}$ denotes MP, AMP, and ISTA.

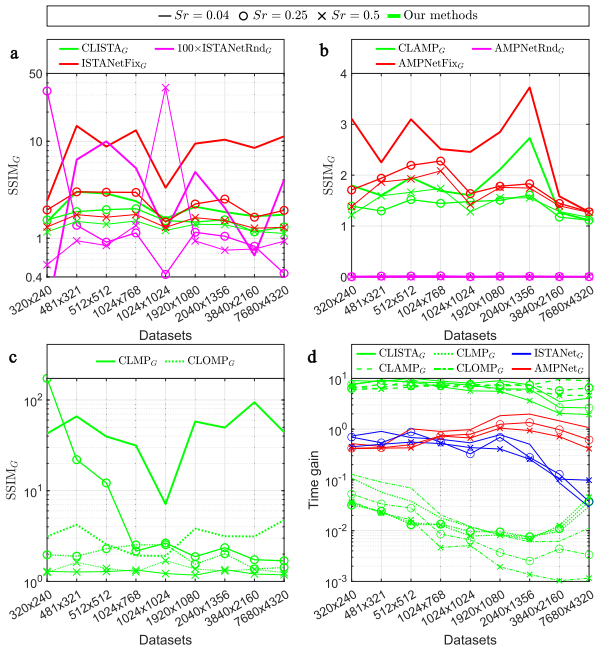


Fig. 13. The propose CL methods versus other methods. (**a**-**c**) The accuracy gain in SSIM (denoted as SSIM$_G$) for the convex optimization method set (using ISTA), the message-passing set (using AMP), and the greedy algorithm set (using OMP and MP), respectively. (**d**) The corresponding time ratio. All gains and ratios are calculated relative to their respective baseline iterative methods, i.e., ISTA, AMP, OMP, or MP.

and computed in parallel on the same GPU platform. The results are shown in Fig. 13.

To date, there are no Deep Unfolding methods based on greedy algorithms, hence OMPNet and MPNet are not shown in the figure. Fig. 13(a)–(c) show that across images ranging from low resolution (320 × 240) to 8K (7680 × 4320), the CL methods consistently achieve higher accuracy than their corresponding baseline traditional iterative methods under the same conditions, as indicated by SSIM$_G$ values greater than 1. This improvement stems from the prior information constraints introduced by the CL framework and its ability to suppress coefficients at support positions that might be incorrectly identified by traditional methods. Under the fixed, pre-trained sampling matrices used by Deep Unfolding, its performance surpasses the proposed CL methods due to end-to-end training on large-scale data. However, its performance degrades significantly when switched to random sampling matrices, highlighting its lack of generality. On the nine diverse image datasets tested, CL improves median reconstruction accuracy by approximately 163%, 78%, and 35% at sampling rates of 0.04, 0.25, and 0.5, respectively, compared to classic iterative methods. Fig. 13(d) reveals that for greedy algorithms like MP and OMP, the CL methods offer substantial efficiency gains over their traditional iterative counterparts. In contrast, for convex optimization and message-passing algorithms, some efficiency is traded for accuracy. This is because CL avoids the computationally expensive matrix pseudo-inversion required in greedy algorithms, while it introduces the additional computational burden of the $\mathbf{Ds}$ loss calculation. Although this adds computational cost in some cases, the processing time for CL methods remains below 1 second for images up to 4K resolution. As shown similarly for CLOMP in Fig. 10, the time is typically around 0.2 seconds, which meets the requirements of most practical applications. Furthermore, while Deep Unfolding methods process images in small patches, our method processes the image globally. Implementing a parallel, patch-based processing scheme for CL would likely further reduce its runtime significantly. However, such patch-based parallel processing involves tackling issues like blocking artifacts, which falls outside the core focus of this work on general sparse reconstruction. Therefore, tests on a patch-parallel version of CL are not included. As image resolution increases, the computational costs related to $\mathbf{D}$ and $\mathbf{s}$ scale similarly for both CL and traditional iterative methods, so their relative time performance advantages remain unchanged.

Tests on both 1D and 2D data demonstrate that Deep Unfolding methods fail when the sampling matrix differs from the one used in pre-training. Only the CL method possesses the same level of generality as traditional iterative methods, namely the out-of-box capability to adapt to sampling matrices of any size or value, and any sampling rate, without modification.

2) Ablation Study: This section conducts ablation studies for the proposed CL method on real 1D and 2D signal datasets. This paper proposes two general prior terms, TV and LV. The ablation results are presented in Table IV. Overall, removing either TV or LV leads to a decline in accuracy across all sampling rates on both 1D and 2D datasets. Specifically, when TV is removed for 1D data or when both TV and LV are removed for 2D data, the prior information in the CL method becomes equivalent to that of traditional iterative methods, resulting in a clear performance drop. For 2D datasets, the performance degradation from removing both terms is consistently greater than from removing either one individually, which validates the effectiveness of both priors.

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TABLE IV
ABLATION STUDY ON REAL 1D AND 2D SIGANL DATASETS

Datasets	WSN or Facial IR	ECG or BSD68	EEG or SET11	Urban100	Brain MRI	FHD	QHD	4K UHD	8K FUHD
Methods				Degradation of SSIM/MSE/PSNR (%) under		Sr=0.04 Sr=0.25 Sr=0.5			
CLISTA	2.67/1.68/1.27	2.68/10.09/3.26	12.60/2.70/2.51						
-TV	94.35/5413.43/68.55	33.11/776.21/34.89	91.44/18712.13/81.31						
on 1D	69.91/8466.78/58.80	2.28/66.93/7.75	60.05/14559.39/56.09						
CLISTA	80.56/7.76/3.47	59.38/0.55/0.15	53.68/0.09/0.00	56.09/0.76/0.53	88.57/13.15/6.35	64.13/2.84/1.61	69.04/3.70/2.15	73.09/5.97/3.21	73.42/5.14/4.68
-TV	34.30/367.21/40.29	70.73/467.13/47.70	76.80/693.25/55.16	69.08/560.64/52.28	40.64/875.80/43.37	78.62/833.12/54.30	77.30/1148.61/55.78	83.34/1353.85/57.45	86.19/2704.01/66.91
on 2D	16.92/47.89/18.91	44.44/57.01/18.00	28.85/28.58/10.50	14.29/142.18/6.34	9.24/114.02/16.98	4.83/87.92/10.24	7.34/430.41/21.88	36.04/701.68/31.32	72.26/7101.07/56.76
CLISTA	89.45/9.56/4.38	67.84/1.89/1.12	62.00/2.94/1.87	62.80/1.44/0.74	92.24/18.49/9.00	72.22/1.61/1.06	76.16/2.81/1.96	80.07/6.34/3.76	77.58/3.33/4.49
-LV	77.93/526.74/47.32	86.01/583.43/52.64	86.82/802.90/58.41	82.74/657.97/56.26	79.40/1120.38/48.61	87.87/947.37/56.92	87.28/1280.98/58.08	90.62/1515.89/59.62	90.92/2836.93/68.24
on 2D	32.94/1190.13/41.70	71.54/897.54/46.73	78.79/1390.91/55.28	74.74/1228.13/54.36	59.78/2176.07/46.66	79.20/1341.55/53.55	81.97/2033.60/56.43	86.48/2550.61/57.97	91.53/9531.66/69.51
CLISTA	90.33/10.27/4.72	69.22/2.07/1.17	63.51/3.25/2.07	64.21/1.66/0.83	92.81/20.13/9.71	73.45/1.69/1.12	77.36/2.93/2.10	81.19/6.83/4.08	78.62/3.10/4.60
-TV-LV	88.97/578.68/49.19	90.16/615.10/53.87	90.55/827.29/59.07	87.66/681.64/57.14	91.32/1206.69/50.28	91.50/981.84/57.68	91.20/1322.22/58.79	93.49/1570.20/60.33	93.35/2866.73/68.61
on 2D	47.44/2396.96/53.85	83.07/1416.71/55.23	88.15/1999.37/62.12	85.04/1681.26/60.89	67.51/3020.08/52.11	86.48/1788.18/58.84	88.09/2610.33/60.92	90.24/3131.11/61.42	93.08/10443.19/71.63
CLAMP	81.20/985.47/52.48	3.27/35.15/7.83	86.23/1256.63/66.10						
-TV	53.95/1045.37/41.53	18.30/197.37/19.75	13.94/191.25/16.29						
on 1D	1.36/6.37/1.06	0.46/0.65/0.40	0.10/4.02/0.64						
CLAMP	3.72/13.13/3.64	0.29/8.86/2.56	1.76/2.74/1.38	0.76/1.18/0.64	0.42/1.40/0.90	0.85/5.19/2.04	1.33/1.34/0.23	0.99/4.07/0.81	0.89/7.72/0.66
-TV	10.44/149.70/12.51	5.92/4.73/2.39	25.95/88.64/17.53	25.79/176.59/21.99	18.65/463.17/21.33	58.23/268.89/36.13	48.38/172.35/28.91	27.19/24.69/9.67	32.69/9.08/7.57
on 2D	0.66/1.70/0.33	1.53/21.84/0.80	0.55/3.50/1.79	0.97/6.42/1.19	3.97/2.65/2.39	0.55/0.11/0.46	0.40/2.52/0.42	1.45/10.69/4.14	0.34/11.94/0.49
CLAMP	2.62/45.24/15.54	19.00/4.84/1.84	14.09/9.45/4.93	21.07/4.58/2.16	4.45/20.40/5.91	12.94/6.64/0.65	21.89/8.66/1.71	9.33/18.57/7.10	9.78/15.76/5.53
-LV	68.32/2143.18/49.64	70.42/330.67/40.74	76.35/659.92/47.71	70.48/498.89/42.80	66.95/2138.79/46.46	61.30/190.64/30.54	50.17/82.86/16.68	58.78/41.51/14.56	57.92/14.21/11.40
on 2D	0.39/5.87/1.28	13.05/50.43/5.96	11.20/26.60/5.19	16.60/59.02/8.02	1.22/38.43/2.69	11.39/51.37/6.69	15.87/87.17/9.20	8.11/92.69/4.77	30.62/2574.54/27.96
CLAMP	92.95/446.52/48.10	95.40/603.21/54.92	95.07/593.87/57.03	94.44/690.09/59.74	94.46/543.30/42.98	95.09/1109.06/60.51	94.96/1045.21/59.31	96.10/1129.51/59.67	96.97/4458.96/70.66
-TV-LV	75.84/2951.02/54.43	79.36/423.64/46.13	84.37/949.22/54.55	79.18/680.73/49.20	76.40/3825.37/53.97	69.82/290.06/37.29	73.79/221.59/33.95	64.21/64.46/20.10	66.39/24.72/16.77
on 2D	8.11/91.25/6.79	20.87/104.96/10.38	19.89/104.26/13.22	25.01/133.67/14.30	1.31/75.56/5.76	16.96/100.01/9.92	21.37/146.61/13.19	11.65/143.27/7.73	49.10/5049.15/40.23
CLMP	18.07/72.35/34.73	0.73/2.86/1.22	31.05/50.78/10.51						
-TV	12.63/17.21/3.61	9.59/41.43/9.53	17.74/75.69/15.15						
on 1D	14.37/98.61/15.44	10.78/205.81/21.86	12.97/351.56/28.33						
CLMP	17.91/29.25/12.83	13.65/58.37/10.00	42.96/3.61/2.16	42.92/9.70/4.81	55.90/17.39/4.64	71.64/417.92/58.04	14.08/337.78/32.99	42.71/0.21/0.38	24.90/5.64/1.93
-TV	2.26/31.02/5.31	20.79/358.59/16.86	6.78/18.77/3.57	6.23/18.20/3.01	1.58/18.46/2.63	4.10/14.96/1.77	3.87/15.98/2.45	2.16/17.94/0.70	1.28/11.44/0.95
on 2D	3.79/17.43/3.01	0.98/2.21/0.22	0.97/6.16/0.72	0.33/4.65/1.43	0.19/2.27/0.44	0.33/2.73/0.37	0.67/2.50/0.59	0.54/0.66/0.95	0.31/3.68/1.02
CLMP	5.91/3.24/2.40	22.74/13.91/0.69	19.07/22.57/6.80	17.02/18.05/6.33	19.78/20.27/4.81	34.20/235.32/28.51	47.37/244.95/25.53	21.05/9.14/2.26	17.87/13.77/4.18
-LV	25.48/68.10/9.02	24.22/37.08/8.43	28.58/31.05/7.15	28.27/33.66/6.87	29.19/97.52/11.16	22.42/21.39/4.63	30.38/38.53/7.35	19.93/50.05/6.44	19.43/45.45/6.48
on 2D	6.95/24.65/3.51	10.46/17.98/3.41	8.12/7.75/1.42	13.91/14.88/1.86	0.62/13.49/3.06	5.21/3.93/0.86	12.10/12.42/1.89	7.08/20.47/1.34	5.78/3.24/0.86
CLMP	87.43/60.31/22.82	87.68/124.16/29.28	75.60/98.01/25.11	55.03/52.39/15.49	51.50/57.95/13.58	68.35/133.51/29.67	76.24/437.65/42.82	38.62/22.55/6.35	33.39/30.36/7.16
-TV-LV	51.86/252.49/23.11	53.33/109.98/18.42	51.03/99.04/17.10	48.63/85.90/15.84	52.33/270.58/21.92	49.53/77.16/14.26	52.79/105.72/16.18	50.99/129.01/20.16	53.00/181.52/19.67
on 2D	46.70/479.08/27.10	48.79/161.07/21.75	40.31/120.71/18.21	40.71/101.73/15.96	42.29/394.23/23.03	44.95/99.29/16.25	46.52/139.65/17.67	53.33/190.22/23.49	56.38/286.49/23.24

TABLE V
A COMPARISON OF THE OUT-OF-THE-BOX CAPABILITIES AMONG VARIOUS METHODS

Methods	Traditional iteration	Deep unfolding	Coefficient learning (proposed)
Capabilities	OMP,MP, AMP,ISTA	ISTANet, AMPNet	CLOMP,CLMP, CLAMP,CLISTA
Any signal size	✓	×	✓
Any image size	✓	✓	✓
Any measurement matrix	✓	×	✓
Any signal dimension	✓	×	✓
Any sampling rate	✓	×	✓
Learnable weights*	×	171090, 219861	1089
Pre-training time	×	super long	×
Pre-training data	×	massive	×
Interpretability	perfect	middle	perfect
Accuracy	low	high	middle
Time	middle	low or middle	low,low,high,high

* based on an image patch with size 33×33 in ISTANet and AMPNet.

3) Comprehensive Analysis: This section provides a comprehensive comparison of the proposed CL method, traditional iterative methods, and Deep Unfolding methods. The results are summarized in Table V. While Deep Unfolding achieves the highest accuracy, it requires retraining when the sampling rate or sampling matrix changes and cannot offer out-of-the-box support. Moreover, in many applications, collecting largescale retraining data is impractical, and the retraining process itself is far more timeconsuming than running traditional iterative methods. Overall, the proposed CL method outperforms traditional approaches in all aspects except for some potential efficiency disadvantages in convex optimization and messagepassing algorithms. Therefore, it can truly serve as a replacement for them.

The proposed CL method in this paper is implemented using PaddlePaddle or PyTorch. These AI development frameworks already support deployment across multiple platforms. For example, PaddlePaddle can be deployed on ten hardware platforms, including GPU and CPU. This effectively ensures the platform adaptability of both the proposed method and Deep Unfolding during deployment.

Regarding computational requirements, following the core concept of CL, the method can be regarded as an extremely small model trained on a single signal. This represents a much lower demand than Deep Unfolding methods trained even on a single image. As shown in Table V, under comparable conditions, its learnable parameter count is three orders of magnitude lower than that of Deep Unfolding. Since Deep Unfolding, with its significantly larger parameter number, can be trained successfully on datasets containing tens of thousands of images, the CL method can naturally operate without failure.

V. CONCLUSIONS AND LIMITATIONS

This paper proposes a framework called coefficients learning, which consists of a residual injection branch and a s forward branch to substitute tradition iterative methods. The residual injection branch transforms the corresponding component in the iterative algorithm by utilizing fixed parameters in neural layers and activation functions. If the branch is not overly complex, transforming it into GPU implementation is also an efficient choice. For the second block, we directly set s as learnable parameters to bypass the time-consuming equation. We propose several branch structures for s learning. The advantage of using

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a learnable $\mathbf{s}$ is that prior information can be integrated into the loss function to enhance performance. We proposes the total variation and local variation as two universal prior loss terms for both 1D and high-dimensional signals. Another significant advantage for CL is the number of learnable parameters is extremely low compared to other AI method, because only n parameters are required in a minimal structure, resulting in a rapid learning process.

There are also some limitations that worth to be addressed in future work. First, the weight factors for prior information are empirically set currently, a bad setting can decrease accuracy. We find these weight factors are linked to Kr, Sr and signal features. A high Sr needs lower priori weight. An image with smooth regions necessitates a higher weight. However, a quantitative guiding rule has not yet been established. Second, the TV and LV loss needs to calculate $\mathbf{D}s$, when n is large, $\mathbf{D} \in \mathbb{R}^{n\times n}$ will occupy too many memory space, which limits efficiency. This explains why the efficiency decreases, and consequently, the time required for CL increases as n grows.

REFERENCES

[1] "The big question," *Nature Mach. Intell.*, vol. 3, no. 9, p. 737, 2021, doi: 10.1038/s42256-021-00395-y. [Online]. Available: https://www.nature.com/articles/s42256-021-00395-y#citeas

[2] Y. Liu et al., "Sora: A review on background, technology, limitations, and opportunities of large vision models," 2024, *arXiv:2402.17177*.

[3] E. J. Candes and T. Tao, "Near-optimal signal recovery from random projections: Universal encoding strategies?," *IEEE Trans. Inf. Theory*, vol. 52, no. 12, pp. 5406–5425, Dec. 2006.

[4] D. L. Donoho, "Compressed sensing," *IEEE Trans. Inf. Theory*, vol. 52, no. 4, pp. 1289–1306, Apr. 2006.

[5] L. Gao, J. Liang, C. Li, and L. V. Wang, "Single-shot compressed ultrafast photography at one hundred billion frames per second," *Nature*, vol. 516, no. 7529, pp. 74–77, 2014.

[6] E. Hahamovich, S. Monin, Y. Hazan, and A. Rosenthal, "Single pixel imaging at megahertz switching rates via cyclic hadamard masks," *Nature Commun.*, vol. 12, no. 1, 2021, Art. no. 4516.

[7] J. Hunt et al., "Metamaterial apertures for computational imaging," *Science*, vol. 339, no. 6117, pp. 310–313, 2013.

[8] M.-J. Sun et al., "Single-pixel three-dimensional imaging with time-based depth resolution," *Nature Commun.*, vol. 7, no. 1, 2016, Art. no. 12010.

[9] Y. Xu, L. Lu, V. Saragadam, and K. F. Kelly, "A compressive hyperspectral video imaging system using a single-pixel detector," *Nature Commun.*, vol. 15, no. 1, 2024, Art. no. 1456.

[10] L. Zhu, W. Zhang, D. Elnatan, and B. Huang, "Faster using compressed sensing," *Nature Methods*, vol. 9, no. 7, pp. 721–723, 2012.

[11] R. Yao, M. Ochoa, P. Yan, and X. Intes, "Net-FLICS: Fast quantitative wide-field fluorescence lifetime imaging with compressed sensing–A deep learning approach," *Light: Sci. Appl.*, vol. 8, no. 1, 2019, Art. no. 26.

[12] E. Hassan and A. Ghaffari, "ISRSLO: Compressed sensing with image smoothness regularized-smoothed ℓ_0-norm," *Sci. Rep.*, vol. 14, no. 1, 2024, Art. no. 24305.

[13] B. Cleary et al., "Compressed sensing for highly efficient imaging transcriptomics," *Nature Biotechnol.*, vol. 39, no. 8, pp. 936–942, 2021.

[14] K. AraiK et al., "Fourier magnetic imaging with nanoscale resolution and compressed sensing speed-up using electronic spins in diamond," *Nature Nanotechnol.*, vol. 10, no. 10, pp. 859–864, 2015.

[15] R. I. Stantchev et al., "Noninvasive, near-field terahertz imaging of hidden objects using a single-pixel detector," *Sci. Adv.*, vol. 2, no. 6, 2016, Art. no. e1600190.

[16] A. Kyrillidis, A. Kalev, D. Park, S. Bhojanapalli, C. Caramanis, and S. Sanghavi, "Provable compressed sensing quantum state tomography via non-convex methods," *npj Quantum Inf.*, vol. 4, no. 1, 2018, Art. no. 36.

[17] S. Wang, Y. Luo, P. Zuo, L. Pan, Y. Li, and Z. Sun, "In-memory analog solution of compressed sensing recovery in one step," *Sci. Adv.*, vol. 9, no. 50, 2023, Art. no. eadj2908.

[18] A. L. Machidon and V. Pejović, "Deep learning for compressive sensing: A ubiquitous systems perspective," *Artif. Intell. Rev.*, vol. 56, no. 4, pp. 3619–3658, 2023.

[19] E. C. Marques, N. Maciel, L. Naviner, H. Cai, and J. Yang, "A review of sparse recovery algorithms," *IEEE Access*, vol. 7, pp. 1300–1322, 2019.

[20] M. Rani, S. B. Dhok, and R. B. Deshmukh, "A systematic review of compressive sensing: Concepts, implementations and applications," *IEEE Access*, vol. 6, pp. 4875–4894, 2018.

[21] J. Tachella, D. Chen, and M. Davies, "Sensing theorems for unsupervised learning in linear inverse problems," *J. Mach. Learn. Res.*, vol. 24, 2023, Art. no. 39.

[22] K. Wei, A. Aviles-Rivero, J. Liang, Y. Fu, H. Huang, and C. B. Schnlieb, "TFPnP: Tuning-free plug-and-play proximal algorithms with applications to inverse imaging problems," *J. Mach. Learn. Res.*, vol. 23, 2022, Art. no. 16.

[23] B. Zhu, J. Z. Liu, S. F. Cauley, B. R. Rosen, and M. S. Rosen, "Image reconstruction by domain-transform manifold learning," *Nature*, vol. 555, no. 7697, pp. 487–492, 2018.

[24] C. Dehner, G. Zahnd, V. Ntziachristos, and D. Jüstel, "A deep neural network for real-time optoacoustic image reconstruction with adjustable speed of sound," *Nature Mach. Intell.*, vol. 5, no. 10, pp. 1130–1141, 2023.

[25] M. Z. Darestani, A. S. Chaudhari, and R. Heckel, "Measuring robustness in deep learning based compressive sensing," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 2433–2444.

[26] D.-I. Eun, R.W. S. Jang, H. Ha, S. C. L. Jung, and N. Kim, "Deep-learning-based image quality enhancement of compressed sensing magnetic resonance imaging of vessel wall: Comparison of self-supervised and unsupervised approaches," *Sci. Rep.*, vol. 10, no. 1, 2020, Art. no. 13950.

[27] N. Davoudi, X. L. De án-Ben, and D. Razansky, "Deep learning optoacoustic tomography with sparse data," *Nature Mach. Intell.*, vol. 1, no. 10, pp. 453–460, 2019.

[28] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, 2015, Art. no. 436.

[29] K. Zeng, Y. Yang, G. Xiao, and Z. Chen, "A very deep densely connected network for compressed sensing," *IEEE Access*, vol. 7, pp. 85430–85439, 2019.

[30] P. Chen, H. Song, Y. Zeng, X. Guo, and C. Tang, "A real-time and robust neural network model for low-measurement-rate compressed-sensing image reconstruction," *Entropy*, vol. 25, no. 12, 2023, Art. no. 1648.

[31] X. Du, K. Liang, Y. Lv, and S. Qiu, "Fast reconstruction of signal compression sensing based on deep learning," *Sci. Rep.*, vol. 14, no. 1, 2024, Art. no. 5087.

[32] K. Zhang, S. Tang, V. Zhu, M. Barchini, and W. Yang, "An end-to-end recurrent compressed sensing method to denoise, detect and demix calcium imaging data," *Nature Mach. Intell.*, vol. 6, no. 9, pp. 1106–1118, 2024.

[33] H. Zhang, W. Zhang, L. Yu, and G. Bi, "Distributed compressive sensing via LSTM-aided sparse Bayesian learning," *Signal Process.*, vol. 176, 2020, Art. no. 107656.

[34] A. Güemes, C. S. Vila, and S. Discetti, "Super-resolution generative adversarial networks of randomly-seeded fields," *Nature Mach. Intell.*, vol. 4, no. 12, pp. 1165–1173, 2022.

[35] T. M. Quan, T. Nguyen-Duc, and W. K. Jeong, "Compressed sensing MRI reconstruction using a generative adversarial network with a cyclic loss," *IEEE Trans. Med. Imag.*, vol. 37, no. 6, pp. 1488–1497, Jun. 2018.

[36] C. Mou and J. Zhang, "TransCL: Transformer makes strong and flexible compressive learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 5236–5251, Apr. 2023.

[37] M. Genzel, J. Macdonald, and M. März, "Solving inverse problems with deep neural networks–robustness included?," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 1119–1134, Jan. 2023.

[38] K. Kulkarni, S. Lohit, P. Turaga, R. Kerviche, and A. Ashok, "ReconNet: Non-iterative reconstruction of images from compressively sensed measurements," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 449–458.

[39] J. Sun, H. Li, and Z. Xu, "Deep ADMM-Net for compressive sensing MRI," in *Proc. 30th Int. Conf. Neural Inf. Process. Syst.*, 2016, pp. 10–18.

[40] B. Chen and J. Zhang, "Practical compact deep compressed sensing," *IEEE Trans. Pattern Anal. Mach. Intell*, vol. 47, no. 3, pp. 1610–1626, Mar. 2025.

[41] J. Hu, K. Niu, Y. Wang, Y. Zhang, and X. Liu, "Research on deep unfolding network reconstruction method based on scalable sampling of transient signals," *Sci. Rep.*, vol. 14, 2024, Art. no. 27733.

[42] Z. He et al., "A deep unrolled neural network for real-time MRI-guided brain intervention," *Nature Commun.*, vol. 14, no. 1, 2023, Art. no. 8257.

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[43] J. Zhang and B. Ghanem, "ISTA-Net: Interpretable optimization-inspired deep network for image compressive sensing," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 1828–1837.
[44] Y. Yang, J. Sun, H. Li, and Z. Xu, "ADMM-CSNet: A deep learning approach for image compressive sensing," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 3, pp. 521–538, Mar. 2020.
[45] Z. Zhang, Y. Liu, J. Liu, F. Wen, and C. Zhu, "AMP-Net: Denoising-based deep unfolding for compressive image sensing," *IEEE Trans. Image Process.*, vol. 30, pp. 1487–1500, 2021.
[46] S.-W. Hu, G.-X. Lin, and C.-S. Lu, "GPX-ADMM-Net: Interpretable deep neural network for image compressive sensing," *IEEE Access*, vol. 9, pp. 158695–158709, 2021.
[47] D. You, J. Xie, and J. Zhang, "ISTA-Net++: Flexible deep unfolding network for compressive sensing," in *Proc. IEEE Int. Conf. Multimedia Expo*, 2021, pp. 1–6.
[48] Z. Fang, "A high-efficient hybrid physics-informed neural networks based on convolutional neural network," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 10, pp. 5514–5526, Oct. 2022.
[49] G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang, "Physics-informed machine learning," *Nature Rev. Phys.*, vol. 3, no. 6, pp. 422–440, 2021.
[50] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *J. Comput. Phys.*, vol. 378, no. 1, pp. 686–707, 2019.
[51] C. Sun et al., "Integrated microring spectrometer with in-hardware compressed sensing to break the resolution-bandwidth limit for general continuous spectrum analysis," *Laser Photon. Rev.*, vol. 17, no. 11, 2023, Art. no. 2300291.
[52] I. Ali, I. Ahmedy, A. Gani, M. U. Munir, and M. H. Anisi, "Data collection in studies on Internet of Things (IoT), wireless sensor networks (WSNs), and sensor cloud (SC): Similarities and differences," *IEEE Access*, vol. 10, pp. 33909–33931, 2022.



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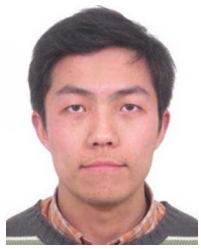
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